

Real Analysis: Part I

Metric Spaces

A concise formal reference for N. L. Carothers

Scope. Chapters 1–9 of the 2000 Cambridge University Press edition, pp. 3–136. Definitions, mathematical statements, identities, reusable exercise results and essential counterexamples are collected here without proofs. Related assertions are grouped; routine numerical computations and repeated definitions are omitted. This edition condenses the existing expanded study reference. It is not a transcription of the textbook.

Source tags. *Th.*, *Prop.*, *Lemma* and *Cor.* refer to textbook statement numbers; *Ex. c.n* refers to exercise n in Chapter c . A range or a list groups related exercise conclusions. *Consequence* marks an auxiliary standard result. These mappings are inherited from the expanded reference; the complete textbook was not available for a fresh, exercise-by-exercise audit.

Numbering. Statement numbers in this file are independent of the book’s numbering. Definitions may also record useful equivalent characterizations. Hypotheses are part of every statement.

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Bibliographic sources.

N. L. Carothers, *Real Analysis*, Cambridge University Press, 2000. doi:10.1017/CBO9780511814228.

Cambridge University Press, [official chapter listing](#).

Supplementary errata reported by Rupert Frank (2013): https://aaron.na31.org/Carothers_Errata.pdf. The Cantor-gap digit statement uses *ternary* expansions; the Hilbert-cube estimate has tail bound 2^{1-k} .

Layout. The source specifies 38 thematic pages. Each page is fitted to its available height so that individual statements are not split. The source has been checked structurally; a TeX compiler was unavailable in the authoring environment, so rendering and font sizes have not been visually verified.

Notation and standing conventions

Fixed throughout the reference

Notation	Meaning
$\mathbb{N}, \mathbb{N}_0, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	$\mathbb{N} = \{1, 2, \dots\}$, $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$; integers, rationals, reals and complex numbers.
$\overline{\mathbb{R}}$	$\mathbb{R} \cup \{-\infty, +\infty\}$. Extended arithmetic is used only when defined.
$\mathbb{F}$	A fixed scalar field in $\{\mathbb{R}, \mathbb{C}\}$ in each normed-space assertion.
$(X, d), (Y, \rho)$	Metric spaces. A subset carries the restricted metric unless specified otherwise.
$B_r^X(x)$	$\{y \in X : d(x, y) < r\}$.
$\overline{B}_r^X(x)$	$\{y \in X : d(x, y) \leq r\}$; this need not equal $\overline{B_r^X(x)}$.
A^c	Complement in the stated ambient space.
$\overline{A}^X, A^\circ, \partial_X A, A'$	Closure, interior, boundary and derived set. Ambient notation is omitted only when unambiguous.
$\mathbf{1}_A$	Characteristic function: $\mathbf{1}_A(x) = 1$ on A , and 0 off A .
$\text{Cont}(f), D(f)$	Continuity and discontinuity sets relative to the domain: $D(f) = X \setminus \text{Cont}(f)$ for $f : X \rightarrow Y$.
$C(X), C(X, \mathbb{F})$	Continuous real functions, or continuous scalar functions when the field is shown.
$C_b(X, \mathbb{F}), B(S)$	Bounded continuous scalar functions; bounded scalar functions on S , both with supremum norm. The book uses $\ell^\infty(S)$ for $B(S)$.
$\mathcal{L}(V, W)$	Bounded linear maps $V \rightarrow W$; the book also uses $B(V, W)$.
$\mathcal{K}$	The ternary Cantor set; denoted Δ in the book.
H, d_H	$H = [-1, 1]^\mathbb{N}$, $d_H(x, y) = \sum_{n \geq 1} 2^{-n} x_n - y_n $; the book writes H^∞ .
$\mathfrak{c}$	$ \mathbb{R} = 2^{\aleph_0}$, the cardinality of the continuum.
$\cong, \sim$	Homeomorphism; equivalence of cardinalities or metrics, with the intended relation specified.
m	Lebesgue measure; used only for length statements about Cantor-type sets and category/measure contrasts.

Quantifiers. Indices n, k, N range over $\mathbb{N}$, unless stated otherwise. For all sufficiently large n means $\exists N \forall n \geq N$; for infinitely many n means $\forall N \exists n \geq N$.

Limits and domains. Scalar functions and sequences are real-valued unless another field is given. Convergence is in the specified metric or norm, with the limit in the specified space. Endpoint continuity is relative to the domain. Increasing and decreasing allow equality; strict monotonicity is explicitly stated.

Intervals and integrals. For a nondegenerate compact interval $[a, b]$, $a < b$. Integrals of continuous functions over such intervals are Riemann integrals; no measure-theoretic integration theory is assumed here.

Empty sets. The empty space is bounded, totally bounded, complete, compact and connected. Set $\text{diam } \emptyset = 0$; the supremum norm on the empty function space is 0. Existence of a fixed point or an attained extremum requires a nonempty domain.

Set theory. Cardinality and countable-selection statements use the usual axiom of choice. A function is constant precisely when its range has at most one point.

1. Calculus Review

Number systems, order, and elementary inequalities

Definition 1.1 (Number systems and set notation).

$$\begin{aligned}\mathbb{N} &= \{1, 2, \dots\}, & \mathbb{Z} &= \{\dots, -2, -1, 0, 1, 2, \dots\}, \\ \mathbb{Q} &= \{m/n : m \in \mathbb{Z}, n \in \mathbb{N}\}, & \mathbb{C} &= \{x + iy : x, y \in \mathbb{R}, i^2 = -1\}, \\ \mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}, & \text{irrational numbers} &= \mathbb{R} \setminus \mathbb{Q}.\end{aligned}$$

For $A, B \subset S$,

$$\begin{aligned}A \setminus B &= \{x \in A : x \notin B\}, & A^c &= S \setminus A, \\ \bigcup_{j \in J} A_j &= \{x : \exists j \in J, x \in A_j\}, & \bigcap_{j \in J} A_j &= \{x : \forall j \in J, x \in A_j\}.\end{aligned}$$

Definition 1.2 (Ordered-field structure). The operations $+, \cdot : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ satisfy, for all $x, y, z \in \mathbb{R}$,

$$\begin{aligned}x + y &= y + x, & (x + y) + z &= x + (y + z), & x + 0 &= x, \\ xy &= yx, & (xy)z &= x(yz), & x \cdot 1 &= x, & 0 \neq 1, \\ \forall x \in \mathbb{R} \exists u \in \mathbb{R} : x + u &= 0, \\ \forall x \in \mathbb{R} \setminus \{0\} \exists v \in \mathbb{R} : xv &= 1, \\ x(y + z) &= xy + xz.\end{aligned}$$

The order satisfies, for all $x, y, z \in \mathbb{R}$,

$$\begin{aligned}x &\leq x, & (x \leq y \wedge y \leq x) &\implies x = y, \\ (x \leq y \wedge y \leq z) &\implies x \leq z, & x \leq y \vee y \leq x, \\ x \leq y &\implies x + z \leq y + z, & (0 \leq x \wedge 0 \leq y) &\implies 0 \leq xy.\end{aligned}$$

Here $x < y \iff x \leq y$ and $x \neq y$. The completeness axiom is stated in Theorem 1.7.

Definition 1.3 (Absolute value and intervals).

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0, \end{cases} \quad [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}, \quad (a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

Half-open intervals are defined by changing the corresponding endpoint inequality. For $a \leq b$, $\text{length}([a, b]) = b - a$.

Lemma 1.4 (Absolute-value identities and inequalities). [Ex. 1.17]

For $x, y \in \mathbb{R}$,

$$\begin{aligned}|x| &\geq 0, & |x| = 0 &\iff x = 0, & |-x| &= |x|, & |xy| &= |x| |y|, \\ |x + y| &\leq |x| + |y|, & ||x| - |y|| &\leq |x - y|, \\ \max\{x, y\} &= \frac{x + y + |x - y|}{2}, & \min\{x, y\} &= \frac{x + y - |x - y|}{2}.\end{aligned}$$

For $\varepsilon > 0$, $|x - y| < \varepsilon \iff y - \varepsilon < x < y + \varepsilon$.

Definition 1.5 (Bounds and boundedness). For $A \subset \mathbb{R}$, define

$$U(A) = \{u \in \mathbb{R} : \forall a \in A, a \leq u\}, \quad L(A) = \{\ell \in \mathbb{R} : \forall a \in A, \ell \leq a\}.$$

$$A \text{ bounded above} \iff U(A) \neq \emptyset,$$

$$A \text{ bounded below} \iff L(A) \neq \emptyset,$$

$$A \text{ bounded} \iff \exists M \geq 0 \forall a \in A : |a| \leq M.$$

1. Calculus Review

Bounds, supremum, infimum, and completeness (continued)

Definition 1.6 (Supremum, infimum, maximum, minimum). For $A \neq \emptyset$,

$$\begin{aligned} s = \sup A \in \mathbb{R} &\iff (\forall a \in A, a \leq s) \wedge (\forall u \in U(A), s \leq u), \\ m = \inf A \in \mathbb{R} &\iff (\forall a \in A, m \leq a) \wedge (\forall \ell \in L(A), \ell \leq m). \end{aligned}$$

$$\max A = s \iff s = \sup A \text{ and } s \in A, \quad \min A = m \iff m = \inf A \text{ and } m \in A.$$

Theorem 1.7 (Least-upper-bound axiom). [\[completeness axiom\]](#)

$$\forall A \subset \mathbb{R} : (A \neq \emptyset \wedge U(A) \neq \emptyset) \implies \exists! s \in \mathbb{R} : s = \sup A.$$

Corollary 1.8 (Greatest lower bounds). [\[Ex. 1.1\]](#)

Every nonempty subset of $\mathbb{R}$ bounded below has a unique infimum. For $-A = \{-a : a \in A\}$, $\inf A = -\sup(-A)$, $\sup A = -\inf(-A)$.

Convention 1.9 (Extended extrema). For arbitrary $A \subset \mathbb{R}$,

$$\begin{aligned} \sup \emptyset &= -\infty, & \inf \emptyset &= +\infty, \\ A \text{ unbounded above} &\implies \sup A = +\infty, \\ A \text{ unbounded below} &\implies \inf A = -\infty. \end{aligned}$$

The order is $-\infty < x < +\infty$ for every $x \in \mathbb{R}$. Expressions such as $+\infty - \infty$ are not defined.

Lemma 1.10 (Approximation of extrema). [\[Ex. 1.3\]](#)

For nonempty $A \subset \mathbb{R}$ and finite s, m ,

$$\begin{aligned} s = \sup A &\iff s \in U(A) \wedge (\forall \varepsilon > 0 \exists a \in A : s - \varepsilon < a \leq s), \\ m = \inf A &\iff m \in L(A) \wedge (\forall \varepsilon > 0 \exists a \in A : m \leq a < m + \varepsilon). \end{aligned}$$

Lemma 1.11 (Comparison and bounds of bounds). [\[Ex. 1.2\]](#)

If $\emptyset \neq B \subset A \subset \mathbb{R}$ and A is bounded, then $-\infty < \inf A \leq \inf B \leq \sup B \leq \sup A < +\infty$.

If A contains two distinct points, then $\inf A < \sup A$. For nonempty bounded A , $U(A) = [\sup A, +\infty)$, $L(A) = (-\infty, \inf A]$, $\inf U(A) = \sup A$, $\sup L(A) = \inf A$.

Lemma 1.12 (Algebra of bounded-set extrema). [\[Consequence\]](#)

Let A, B be nonempty bounded subsets of $\mathbb{R}$, let $c \in \mathbb{R}$, and set $A + B = \{a + b : a \in A, b \in B\}$. Then

$$\begin{aligned} \sup(A \cup B) &= \max\{\sup A, \sup B\}, & \inf(A \cup B) &= \min\{\inf A, \inf B\}, \\ \sup(A + B) &= \sup A + \sup B, & \inf(A + B) &= \inf A + \inf B, \\ \sup(c + A) &= c + \sup A, & \inf(c + A) &= c + \inf A. \end{aligned}$$

For $\lambda > 0$, $\sup(\lambda A) = \lambda \sup A$, $\inf(\lambda A) = \lambda \inf A$;

for $\lambda < 0$, $\sup(\lambda A) = \lambda \inf A$, $\inf(\lambda A) = \lambda \sup A$.

Lemma 1.13 (Archimedean property). [\[Lemma 1.2\]](#)

$$\forall x, y > 0 \exists n \in \mathbb{N} : nx > y.$$

Equivalently,

$$\forall M \in \mathbb{R} \exists n \in \mathbb{N} : n > M, \quad \forall \varepsilon > 0 \exists n \in \mathbb{N} : 0 < 1/n < \varepsilon.$$

Corollary 1.14 (Integer part). [\[Consequence\]](#)

For every $x \in \mathbb{R}$, there is a unique $m \in \mathbb{Z}$ such that $m \leq x < m + 1$.

This integer is $\lfloor x \rfloor$. Also, $\lceil x \rceil = -\lfloor -x \rfloor$.

1. Calculus Review

Archimedean property and density (continued)

Theorem 1.15 (Density of rational and irrational numbers). [Th. 1.3][Ex. 1.7–8]

For every $a < b$,

$$\exists q \in \mathbb{Q} : a < q < b, \quad \exists r \in \mathbb{R} \setminus \mathbb{Q} : a < r < b.$$

Both $(a, b) \cap \mathbb{Q}$ and $(a, b) \cap (\mathbb{R} \setminus \mathbb{Q})$ are infinite.

Corollary 1.16 (Monotone approximation by dense sets). For every $x \in \mathbb{R}$ and each $E \in \{\mathbb{Q}, \mathbb{R} \setminus \mathbb{Q}\}$, there exist sequences $(u_n), (v_n) \subset E$ such that $u_n < u_{n+1} < x < v_{n+1} < v_n$, $u_n \rightarrow x$, $v_n \rightarrow x$.

Example 1.17 (Order completeness of $\mathbb{Z}$ and its failure in $\mathbb{Q}$). [Ex. 1.9]

$$\emptyset \neq A \subset \mathbb{Z}, \quad \exists M \in \mathbb{Z} \forall a \in A : a \leq M \implies \max A \in \mathbb{Z}.$$

In contrast, $A = \{q \in \mathbb{Q} : 0 \leq q, q^2 < 2\}$

is nonempty and bounded above in $\mathbb{Q}$, but has no supremum in $\mathbb{Q}$. Its supremum in $\mathbb{R}$ is $\sqrt{2}$.

Definition 1.18 (Sequences, convergence, boundedness). A real sequence is a map $x : \mathbb{N} \rightarrow \mathbb{R}$, written (x_n) . For $\ell \in \mathbb{R}$,

$$x_n \rightarrow \ell \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N : |x_n - \ell| < \varepsilon.$$

$$(x_n) \text{ bounded} \iff \exists M \geq 0 \forall n \in \mathbb{N} : |x_n| \leq M.$$

Write $\sup_n x_n = \sup\{x_n : n \in \mathbb{N}\}$ and similarly for $\inf_n x_n$.

Definition 1.19 (Eventually and infinitely often). For assertions $P(n)$,

$$P(n) \text{ eventually} \iff \exists N \in \mathbb{N} \forall n \geq N : P(n),$$

$$P(n) \text{ infinitely often} \iff \forall N \in \mathbb{N} \exists n \geq N : P(n).$$

“For all but finitely many n ” means “eventually.”

Definition 1.20 (Extended convergence).

$$x_n \rightarrow +\infty \iff \forall M \in \mathbb{R} \exists N \forall n \geq N : x_n > M,$$

$$x_n \rightarrow -\infty \iff \forall M \in \mathbb{R} \exists N \forall n \geq N : x_n < M.$$

These are extended limits, not convergence to a real number.

Lemma 1.21 (Uniqueness, boundedness, and passage of order to limits). [Ex. 1.5–6]

$$x_n \rightarrow \ell, x_n \rightarrow m \implies \ell = m, \quad x_n \rightarrow \ell \in \mathbb{R} \implies (x_n) \text{ bounded}.$$

If $x_n \rightarrow \ell$, $y_n \rightarrow m$, and $x_n \leq y_n$ eventually, then $\ell \leq m$. In particular, $x_n \leq b$ eventually $\implies \ell \leq b$, $\inf_n x_n \leq \ell \leq \sup_n x_n$.

Strict inequalities need not survive: $b - 1/n < b$, but $b - 1/n \rightarrow b$.

Lemma 1.22 (Finite limit laws). [Consequence]

If $x_n \rightarrow x$, $y_n \rightarrow y$, and $\alpha, \beta \in \mathbb{R}$, then

$$\begin{aligned} \alpha x_n + \beta y_n &\rightarrow \alpha x + \beta y, & x_n y_n &\rightarrow xy, & |x_n| &\rightarrow |x|, \\ \min\{x_n, y_n\} &\rightarrow \min\{x, y\}, & \max\{x_n, y_n\} &\rightarrow \max\{x, y\}. \end{aligned}$$

If $y \neq 0$, then $y_n \neq 0$ eventually and $x_n/y_n \rightarrow x/y$. If $x_n \rightarrow 0$ and (y_n) is bounded, then $x_n y_n \rightarrow 0$.

Lemma 1.23 (Squeeze principle). [Consequence]

$$u_n \leq x_n \leq v_n \text{ eventually}, \quad u_n \rightarrow \ell, v_n \rightarrow \ell \implies x_n \rightarrow \ell.$$

Changing or deleting finitely many terms does not change a limit.

Definition 1.24 (Subsequence). A subsequence is $(x_{n_k})_{k \in \mathbb{N}}$, where $n_1 < n_2 < \dots$, $n_k \in \mathbb{N}$.

In particular, $n_k \geq k$ and $n_k \rightarrow \infty$.

Lemma 1.25 (Subsequences and interleaving). [Consequence]

If $x_n \rightarrow \ell \in \mathbb{R}$, then every subsequence converges to ℓ . For real sequences $(u_n), (v_n)$, set $z_{2n-1} = u_n$, $z_{2n} = v_n$. Then $z_n \rightarrow \ell \iff u_n \rightarrow \ell$ and $v_n \rightarrow \ell$.

1. Calculus Review

Monotone convergence and nested intervals (continued)

Definition 1.26 (Monotone sequences).

$$(x_n) \text{ increasing} \iff x_n \leq x_{n+1} \quad \forall n, \quad (x_n) \text{ decreasing} \iff x_{n+1} \leq x_n \quad \forall n.$$

A sequence is monotone if it satisfies at least one of these conditions.

Theorem 1.27 (Monotone convergence). [Th. 1.4]

$$\begin{aligned} (x_n) \text{ increasing and bounded above} &\implies x_n \rightarrow \sup_n x_n \in \mathbb{R}, \\ (x_n) \text{ decreasing and bounded below} &\implies x_n \rightarrow \inf_n x_n \in \mathbb{R}. \end{aligned}$$

Without boundedness, the corresponding extended limits are $+\infty$ and $-\infty$. Thus every monotone real sequence has an extended limit.

Theorem 1.28 (Nested interval theorem). [Th. 1.5]

Let $I_n = [a_n, b_n]$ with finite endpoints, $a_n \leq b_n$, and $I_{n+1} \subset I_n$. Then $a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$, $\bigcap_{n=1}^{\infty} I_n = [\alpha, \beta] \neq \emptyset$, where $\alpha = \sup_n a_n = \lim_n a_n$, $\beta = \inf_n b_n = \lim_n b_n$, $\alpha \leq \beta$.

If $b_n - a_n \rightarrow 0$, then $\alpha = \beta$ and $\bigcap_n I_n = \{\alpha\}$.

Corollary 1.29 (Arbitrary selections from shrinking intervals). If the intervals above satisfy $b_n - a_n \rightarrow 0$ and $x_n \in I_n$, then $|x_n - \alpha| \leq b_n - a_n \rightarrow 0$.

Example 1.30 (Necessity of the interval hypotheses). $\bigcap_{n=1}^{\infty} [n, +\infty) = \emptyset$, $\bigcap_{n=1}^{\infty} (0, 1/n] = \emptyset$.

The first family consists of unbounded closed intervals; the second consists of bounded intervals that are not closed.

Definition 1.31 (Cauchy sequence).

$$(x_n) \text{ Cauchy} \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N : |x_n - x_m| < \varepsilon.$$

Lemma 1.32 (Basic Cauchy properties). [Ex. 1.14–15]

$$\begin{aligned} x_n \rightarrow \ell \in \mathbb{R} &\implies (x_n) \text{ Cauchy}, \\ (x_n) \text{ Cauchy} &\implies (x_n) \text{ bounded}, \\ (x_n) \text{ Cauchy}, \quad x_{n_k} \rightarrow \ell \in \mathbb{R} &\implies x_n \rightarrow \ell. \end{aligned}$$

Every subsequence of a Cauchy sequence is Cauchy.

Theorem 1.33 (Sequential completeness of $\mathbb{R}$). [Cor. 1.13; Th. 1.16]

$$(x_n) \text{ Cauchy} \iff \exists \ell \in \mathbb{R} : x_n \rightarrow \ell.$$

Lemma 1.34 (Small successive differences are insufficient). [Consequence]

A Cauchy sequence satisfies $x_{n+1} - x_n \rightarrow 0$. The converse fails:

$$x_n = \sqrt{n}, \quad x_{n+1} - x_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} \rightarrow 0, \quad x_n \rightarrow +\infty.$$

Theorem 1.35 (Equivalent forms of completeness). [Consequence]

For an Archimedean ordered field F , the following are equivalent (bounds, limits, and Cauchy conditions are taken in F):

- (i) Every nonempty subset of F bounded above has a supremum in F .
- (ii) Every increasing sequence bounded above converges in F .
- (iii) Every nested sequence of nonempty closed bounded intervals in F has nonempty intersection.
- (iv) Every bounded sequence in F has a subsequence converging in F .
- (v) Every Cauchy sequence in F converges in F .

All five hold for $F = \mathbb{R}$. The Archimedean hypothesis is retained in this general-field formulation.

1. Calculus Review

Series, inequalities, and elementary iterative sequences (continued)

Definition 1.36 (Infinite series). For a real sequence (a_n) , its partial sums are $s_N = \sum_{n=1}^N a_n$. Define

$$\sum_{n=1}^{\infty} a_n = s \in \mathbb{R} \iff s_N \rightarrow s.$$

Absolute convergence means $\sum_n |a_n| < \infty$.

Lemma 1.37 (Nonnegative series and the Cauchy criterion). [Ex. 1.13][Consequence]

If $a_n \geq 0$, then

$$\sum_{n=1}^{\infty} a_n = \sup_{N \in \mathbb{N}} \sum_{n=1}^N a_n \in [0, +\infty],$$

and the series is finite exactly when its partial sums are bounded above. For arbitrary real a_n ,

$$\sum_n a_n \text{ converges} \iff \forall \varepsilon > 0 \exists N \forall m \geq n \geq N : \left| \sum_{k=n}^m a_k \right| < \varepsilon.$$

Corollary 1.38 (Basic series consequences). [Consequence]

$$\begin{aligned} \sum_n a_n \text{ converges} &\implies a_n \rightarrow 0, \\ 0 \leq a_n \leq b_n \text{ eventually, } \sum_n b_n < \infty &\implies \sum_n a_n < \infty, \\ \sum_n |a_n| < \infty &\implies \sum_n a_n \text{ converges.} \end{aligned}$$

For $|q| < 1$,

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}, \quad \sum_{n=N+1}^{\infty} q^n = \frac{q^{N+1}}{1-q}.$$

Theorem 1.39 (Bernoulli inequality). [Bernoulli's Inequality 1.9][Ex. 1.18(a)]

For $t \geq -1$ and $n \in \mathbb{N}$, $(1+t)^n \geq 1+nt$.

If $t > -1$, $t \neq 0$, and $n \geq 2$, the inequality is strict.

Corollary 1.40 (Real exponents). [Ex. 1.18(c)–(d)]

For $t > 0$ and $r > 1$ real, $(1+t)^r > 1+rt$.

More generally, $(1+t)^r \geq 1+rt$ for $t > -1$, $r \geq 1$, with strict inequality when $t \neq 0$, $r > 1$.

Theorem 1.41 (Arithmetic–geometric mean inequality). [Ex. 1.20]

For $a_1, \dots, a_n \geq 0$,

$$\left(\prod_{j=1}^n a_j \right)^{1/n} \leq \frac{1}{n} \sum_{j=1}^n a_j.$$

Equality holds exactly when $a_1 = \dots = a_n$. In particular, $\sqrt{ab} \leq (a+b)/2$ for $a, b \geq 0$.

Lemma 1.42 (Powers and roots). [Ex. 1.19]

$0 < c < 1 \implies c^n \rightarrow 0$, $c > 0 \implies c^{1/n} \rightarrow 1$.

Consequently, $c > 1 \implies c^n \rightarrow +\infty$.

Proposition 1.43 (Newton iteration for a positive square root). [Ex. 1.11]

Let $a > 0$, $x_1 > \sqrt{a}$, and

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right).$$

Then

$$\sqrt{a} < x_{n+1} < x_n, \quad x_n \downarrow \sqrt{a}, \quad x_{n+1} - \sqrt{a} = \frac{(x_n - \sqrt{a})^2}{2x_n}.$$

The same iteration converges for every $x_1 > 0$: $x_n \geq \sqrt{a}$ for $n \geq 2$, and the sequence is decreasing from $n = 2$ onward.

Definition 1.44 (Base p expansion). Fix an integer $p \geq 2$. A digit sequence satisfies $d_n \in \{0, 1, \dots, p-1\}$. Its represented value is

$$(0.d_1 d_2 \dots)_p := \sum_{n=1}^{\infty} \frac{d_n}{p^n}.$$

It is terminating if $d_n = 0$ eventually, purely periodic if $\exists m \in \mathbb{N} \forall n \in \mathbb{N} : d_{n+m} = d_n$, and eventually periodic if $\exists m, N \in \mathbb{N} \forall n \geq N : d_{n+m} = d_n$.

1. Calculus Review

Base expansions and real powers (continued)

Proposition 1.45 (Convergence and representation). [Props. 1.7–1.8]

Every digit sequence represents a point of $[0, 1]$, and every point of $[0, 1]$ has such a representation. If $S_N = \sum_{n=1}^N d_n/p^n$ represents x , then $0 \leq S_N \leq x \leq 1$, $0 \leq x - S_N \leq p^{-N}$.

For $0 \leq x < 1$, one choice is

$$d_n = \lfloor p^n x \rfloor - p \lfloor p^{n-1} x \rfloor, \quad S_N = \frac{\lfloor p^N x \rfloor}{p^N}, \quad 0 \leq x - S_N < p^{-N}.$$

For $x = 1$, take $d_n = p - 1$ for every n .

Theorem 1.46 (Uniqueness and the two-expansion case). [Ex. 1.16, 21]

For $0 < x < 1$, exactly two base p expansions exist if and only if $x = q/p^N$ for some $N \in \mathbb{N}$, $q \in \{1, \dots, p^N - 1\}$.

Every other $x \in (0, 1)$ has exactly one expansion. If a terminating expansion has last nonzero digit d_N , its other expansion is $(0.d_1 \cdots d_{N-1} d_N 000 \cdots)_p = (0.d_1 \cdots d_{N-1} (d_N - 1)(p - 1)(p - 1) \cdots)_p$.

With the convention $x = (0.d_1 d_2 \cdots)_p$, 0 has only the all-zero expansion and 1 only the all- $(p - 1)$ expansion. Allowing an integer part also permits $1.000 \cdots$.

Theorem 1.47 (Periodic expansions and rationality). [Ex. 1.21]

For $x \in [0, 1]$, $x \in \mathbb{Q} \iff x$ has an eventually periodic base p expansion.

If the repeated block has length m and integer value $B = \sum_{j=1}^m d_j p^{m-j}$, then

$$(0.\overline{d_1 \cdots d_m})_p = \frac{B}{p^m - 1}.$$

If a prefix of length N has integer value A , followed by this block, then

$$x = \frac{A}{p^N} + \frac{B}{p^N(p^m - 1)}.$$

For $x = a/b \in (0, 1)$ in lowest terms,

$$\begin{aligned} x \text{ has a terminating expansion} &\iff \exists N \in \mathbb{N} : b \mid p^N, \\ x \text{ has a purely periodic expansion} &\iff \gcd(b, p) = 1. \end{aligned}$$

Lemma 1.48 (Existence and uniqueness of positive roots). [Consequence]

$$\forall A > 0 \ \forall n \in \mathbb{N} \ \exists! u > 0 : u^n = A.$$

This root is denoted by $A^{1/n}$. For $A = 0$, the unique nonnegative n -th root is 0.

Definition 1.49 (Positive-base real powers). For $A > 0$ and $r = m/n \in \mathbb{Q}$, with $m \in \mathbb{Z}$, $n \in \mathbb{N}$, define $A^r = (A^{1/n})^m$, where $A^{1/n}$ is the unique positive n -th root. For $A > 1$, $x \in \mathbb{R}$, set $A^x = \sup\{A^r : r \in \mathbb{Q}, r < x\}$.

For $0 < A < 1$, set $A^x = (1/A)^{-x}$; set $1^x = 1$.

Lemma 1.50 (Laws of real powers). [Consequence]

For $A, B > 0$, $x, y \in \mathbb{R}$,

$$\begin{aligned} A^0 &= 1, & A^{x+y} &= A^x A^y, & A^{-x} &= 1/A^x, \\ (A^x)^y &= A^{xy}, & (AB)^x &= A^x B^x. \end{aligned}$$

The function $x \mapsto A^x$ is continuous; it is strictly increasing when $A > 1$, strictly decreasing when $0 < A < 1$, and constant when $A = 1$.

Proposition 1.51 (Monotone bounds for e). [Prop. 1.10]

Set

$$A_n = \left(1 + \frac{1}{n}\right)^n, \quad B_n = \left(1 + \frac{1}{n}\right)^{n+1}.$$

Then $A_n < A_{n+1}$, $B_{n+1} < B_n$, $2 \leq A_n < B_n \leq 4$.

Both sequences converge to the same number: $e := \lim_n A_n = \lim_n B_n$, $2 < e < 4$, $A_n < e < B_n$.

In particular,

$$0 < B_n - A_n = \frac{A_n}{n} \leq \frac{4}{n}.$$

1. Calculus Review

Construction of e (continued)

Proposition 1.52 (Exponential limit). [Ex. 1.18(b)]

For $t > 0$, $((1 + t/n)^n)$ is strictly increasing. For every $t \in \mathbb{R}$,

$$e^t = \lim_{n \rightarrow \infty} \left(1 + \frac{t}{n}\right)^n,$$

where n is sufficiently large that $1 + t/n > 0$. Here e^t is the real power defined in Definition 1.49.

Definition 1.53 (Tail extrema, liminf, limsup). For any real sequence (x_n) , define $\alpha_N = \inf_{n \geq N} x_n$, $\beta_N = \sup_{n \geq N} x_n$ in $\overline{\mathbb{R}}$.

$\liminf_{n \rightarrow \infty} x_n = \sup_{N \in \mathbb{N}} \alpha_N$, $\limsup_{n \rightarrow \infty} x_n = \inf_{N \in \mathbb{N}} \beta_N$.

Thus α_N increases, β_N decreases, and their extended limits are the liminf and limsup, respectively.

Lemma 1.54 (Basic bounds and sign reversal). [Ex. 1.22, 24]

$\inf_n x_n \leq \liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n \leq \sup_n x_n$,

$\limsup_{n \rightarrow \infty} (-x_n) = -\liminf_{n \rightarrow \infty} x_n$, $\liminf_{n \rightarrow \infty} (-x_n) = -\limsup_{n \rightarrow \infty} x_n$.

These statements allow infinite values. Both liminf and limsup are unchanged by altering finitely many terms.

Theorem 1.55 (Quantified characterization of finite limsup and liminf). [Ex. 1.26]

For $L \in \mathbb{R}$,

$$\limsup_{n \rightarrow \infty} x_n = L \iff \begin{cases} \forall \varepsilon > 0 \exists N \forall n \geq N : x_n < L + \varepsilon, \\ \forall \varepsilon > 0 \forall N \exists n \geq N : x_n > L - \varepsilon. \end{cases}$$

For $\ell \in \mathbb{R}$,

$$\liminf_{n \rightarrow \infty} x_n = \ell \iff \begin{cases} \forall \varepsilon > 0 \exists N \forall n \geq N : x_n > \ell - \varepsilon, \\ \forall \varepsilon > 0 \forall N \exists n \geq N : x_n < \ell + \varepsilon. \end{cases}$$

Lemma 1.56 (Infinite values). [Ex. 1.25]

$$\limsup_{n \rightarrow \infty} x_n = -\infty \iff x_n \rightarrow -\infty,$$

$$\liminf_{n \rightarrow \infty} x_n = +\infty \iff x_n \rightarrow +\infty,$$

$$\limsup_{n \rightarrow \infty} x_n = +\infty \iff \forall M \in \mathbb{R} \forall N \exists n \geq N : x_n > M,$$

$$\liminf_{n \rightarrow \infty} x_n = -\infty \iff \forall M \in \mathbb{R} \forall N \exists n \geq N : x_n < M.$$

The last two conditions guarantee subsequences tending to the corresponding infinity; they do not require the entire sequence to do so.

Theorem 1.57 (Convergence criterion). [Cor. 1.12; Prop. 1.15][Ex. 1.23]

For $L \in \mathbb{R}$, $x_n \rightarrow L \iff \liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n = L$.

This equivalence also holds with $L = \pm\infty$ under the extended-limit convention. In the bounded case, all three quantities are finite.

Lemma 1.58 (Order and addition). [Ex. 1.30–31][Consequence]

If $x_n \leq y_n$ eventually, then $\liminf_{n \rightarrow \infty} x_n \leq \liminf_{n \rightarrow \infty} y_n$, $\limsup_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} y_n$.

In particular, if $x_n \rightarrow x \in \mathbb{R}$, then $x \leq \liminf_{n \rightarrow \infty} y_n$. For bounded sequences $(x_n), (y_n)$, $\liminf_{n \rightarrow \infty} x_n + \liminf_{n \rightarrow \infty} y_n \leq \liminf_{n \rightarrow \infty} (x_n + y_n) \leq \limsup_{n \rightarrow \infty} (x_n + y_n) \leq \limsup_{n \rightarrow \infty} x_n + \limsup_{n \rightarrow \infty} y_n$.

If $x_n \rightarrow x \in \mathbb{R}$ and (y_n) is bounded, then the sharper equalities hold: $\limsup_{n \rightarrow \infty} (x_n + y_n) = x + \limsup_{n \rightarrow \infty} y_n$, $\liminf_{n \rightarrow \infty} (x_n + y_n) = x + \liminf_{n \rightarrow \infty} y_n$.

The limsup equality strengthens the inequality requested in Exercise 31.

Lemma 1.59 (Scaling). [Consequence]

For bounded (x_n) and $c \in \mathbb{R}$,

$$c > 0 : \limsup_{n \rightarrow \infty} (cx_n) = c \limsup_{n \rightarrow \infty} x_n, \quad \liminf_{n \rightarrow \infty} (cx_n) = c \liminf_{n \rightarrow \infty} x_n,$$

$$c < 0 : \limsup_{n \rightarrow \infty} (cx_n) = c \liminf_{n \rightarrow \infty} x_n, \quad \liminf_{n \rightarrow \infty} (cx_n) = c \limsup_{n \rightarrow \infty} x_n,$$

$$c = 0 : \limsup_{n \rightarrow \infty} (cx_n) = \liminf_{n \rightarrow \infty} (cx_n) = 0.$$

Definition 1.60 (Extended cluster values).

$$\text{Clust}(x_n) = \{L \in \overline{\mathbb{R}} : \exists n_1 < n_2 < \dots, x_{n_k} \rightarrow L\}.$$

1. Calculus Review

Subsequential limits and Bolzano–Weierstrass (continued)

Theorem 1.61 (Extremal cluster values). [Ex. 1.27, 29, 32]

Every real sequence has subsequences converging in $\overline{\mathbb{R}}$ to its liminf and limsup. Moreover, $\emptyset \neq \text{Clust}(x_n)$, $\min \text{Clust}(x_n) = \liminf_{n \rightarrow \infty} x_n$, $\max \text{Clust}(x_n) = \limsup_{n \rightarrow \infty} x_n$.

In particular, $L \in \text{Clust}(x_n) \implies \liminf_{n \rightarrow \infty} x_n \leq L \leq \limsup_{n \rightarrow \infty} x_n$.

For bounded sequences, every cluster value is a real number.

Theorem 1.62 (Monotone subsequence). [Ex. 1.28]

Every real sequence has an increasing or a decreasing subsequence.

Theorem 1.63 (Bolzano–Weierstrass). [Th. 1.11]

$$(x_n) \text{ bounded} \implies \exists L \in \mathbb{R} \exists n_1 < n_2 < \cdots : x_{n_k} \rightarrow L.$$

Corollary 1.64 (Cauchy subsequence). [Cor. 1.14]

Every bounded real sequence has a Cauchy subsequence.

Theorem 1.65 (Subsequence-of-subsequence criterion). [Ex. 1.33]

For a real sequence (x_n) and $L \in \mathbb{R}$,

$$x_n \rightarrow L \iff \text{every subsequence } (x_{n_k}) \text{ has a further subsequence } (x_{n_{k_j}}) \text{ converging to } L.$$

For bounded (x_n) , this is also equivalent to $\text{Clust}(x_n) = \{L\}$.

Lemma 1.66 (Summability and na_n). [Ex. 1.34]

$$a_n \geq 0, \quad \sum_{n=1}^{\infty} a_n < \infty \implies \liminf_{n \rightarrow \infty} na_n = 0.$$

The full limit need not be zero. For example,

$$a_n = \frac{1}{n^2} + \frac{1}{n} \mathbf{1}_{\{2^k : k \in \mathbb{N}\}}(n)$$

is strictly positive and summable, while $\liminf_{n \rightarrow \infty} na_n = 0$, $\limsup_{n \rightarrow \infty} na_n = 1$.

If, additionally, (a_n) is decreasing, then $na_n \rightarrow 0$.

Theorem 1.67 (Ratio test). [Ex. 1.35]

Let $a_n > 0$ and $r_n = a_{n+1}/a_n$. Then

$$\begin{aligned} \limsup_{n \rightarrow \infty} r_n < 1 &\implies \sum_n a_n < \infty, \\ \liminf_{n \rightarrow \infty} r_n > 1 &\implies a_n \rightarrow +\infty \implies \sum_n a_n = +\infty. \end{aligned}$$

The case $r_n \rightarrow 1$ is inconclusive:

$$\sum_n \frac{1}{n^2} < \infty, \quad \sum_n \frac{1}{n} = +\infty,$$

and both term sequences have successive-term ratio tending to 1.

Theorem 1.68 (Root test). [Ex. 1.36]

Let $a_n \geq 0$, and let $\rho = \limsup_{n \rightarrow \infty} a_n^{1/n}$. Then

$$\rho < 1 \implies \sum_n a_n < \infty, \quad \rho > 1 \implies a_n \not\rightarrow 0 \implies \sum_n a_n = +\infty.$$

The second assertion is stronger than the exercise's sufficient condition $\liminf_{n \rightarrow \infty} a_n^{1/n} > 1$. The case $\rho = 1$ is inconclusive; the two series above also have $a_n^{1/n} \rightarrow 1$.

Definition 1.69 (Limsup and liminf of sets). [Ex. 1.37]

For $E_n \subset S$, define

$$\limsup_{n \rightarrow \infty} E_n = \bigcap_{N=1}^{\infty} \bigcup_{n \geq N} E_n, \quad \liminf_{n \rightarrow \infty} E_n = \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} E_n.$$

Lemma 1.70 (Membership and complements). [Ex. 1.37–38]

$$\begin{aligned} x \in \limsup_n E_n &\iff \forall N \exists n \geq N : x \in E_n, \\ x \in \liminf_n E_n &\iff \exists N \forall n \geq N : x \in E_n. \end{aligned}$$

Consequently, $\liminf_n E_n \subset \limsup_n E_n$,

$$\liminf_n (E_n^c) = (\limsup_n E_n)^c, \quad \limsup_n (E_n^c) = (\liminf_n E_n)^c.$$

1. Calculus Review

Limits of sequences of sets (continued)

Definition 1.71 (Set limit). [Ex. 1.39]

$$E_n \rightarrow E \iff \liminf_n E_n = \limsup_n E_n = E.$$

Equivalently, each $x \in S$ belongs to E_n eventually if $x \in E$, and belongs to E_n^c eventually if $x \notin E$.

Lemma 1.72 (Monotone set limits). [Ex. 1.39]

$$E_n \subset E_{n+1} \quad \forall n \implies E_n \rightarrow \bigcup_{n=1}^{\infty} E_n,$$

$$E_{n+1} \subset E_n \quad \forall n \implies E_n \rightarrow \bigcap_{n=1}^{\infty} E_n.$$

No nonemptiness assertion for the decreasing intersection follows without further hypotheses.

Corollary 1.73 (Indicator formulation). [Consequence]

Let $\mathbf{1}_E(x) = 1$ for $x \in E$, and 0 otherwise. For every $x \in S$, $\mathbf{1}_{\limsup_n E_n}(x) = \limsup_{n \rightarrow \infty} \mathbf{1}_{E_n}(x)$, $\mathbf{1}_{\liminf_n E_n}(x) = \liminf_{n \rightarrow \infty} \mathbf{1}_{E_n}(x)$.

Thus

$$E_n \rightarrow E \iff \forall x \in S : \mathbf{1}_{E_n}(x) \rightarrow \mathbf{1}_E(x).$$

Definition 1.74 (Limit relative to a domain). Let $f : D \rightarrow \mathbb{R}$ and let c be an accumulation point of D . For $L \in \mathbb{R}$,

$$\lim_{\substack{x \rightarrow c \\ x \in D}} f(x) = L \iff \forall \varepsilon > 0 \exists \delta > 0 \forall x \in D : 0 < |x - c| < \delta \implies |f(x) - L| < \varepsilon.$$

When D contains a punctured neighborhood of c , write simply $\lim_{x \rightarrow c} f(x) = L$.

Theorem 1.75 (Sequential characterization of a function limit). [Th. 1.17][Ex. 1.40]

Let $f : D \rightarrow \mathbb{R}$, with c an accumulation point of D . The following are equivalent:

- (i) $\lim_{x \rightarrow c, x \in D} f(x)$ exists as a finite real number.
- (ii) There is $L \in \mathbb{R}$ such that for every $(x_n) \subset D \setminus \{c\}$, $x_n \rightarrow c \implies f(x_n) \rightarrow L$.
- (iii) For every $(x_n) \subset D \setminus \{c\}$ converging to c , the sequence $(f(x_n))$ has a finite limit.

In (iii), the limits are necessarily the same for all such sequences. For a specified L , the epsilon–delta definition is equivalent to (ii) with that same L .

Theorem 1.76 (Sequential characterization of continuity). [Th. 1.18][Ex. 1.41]

For $f : D \rightarrow \mathbb{R}$, $c \in D$, the following are equivalent:

- (i) f is continuous at c .
- (ii) For every $(x_n) \subset D$, $x_n \rightarrow c \implies f(x_n) \rightarrow f(c)$.
- (iii) For every $(x_n) \subset D$ converging to c , $(f(x_n))$ has a finite limit.

Condition (iii) forces that limit to equal $f(c)$.

Corollary 1.77 (Sequential tests for failure). [Consequence]

f is not continuous at $c \in D$ if and only if

$$\exists \varepsilon_0 > 0 \exists (x_n) \subset D : x_n \rightarrow c, \quad |f(x_n) - f(c)| \geq \varepsilon_0 \quad \forall n.$$

If two sequences in $D \setminus \{c\}$ approach c and their function values converge to different finite limits, the function has no limit at c .

Definition 1.78 (One-sided limits and one-sided continuity). For a function defined on the corresponding side of c ,

$$f(c+) = L \iff \forall \varepsilon > 0 \exists \delta > 0 \forall x : 0 < x - c < \delta \implies |f(x) - L| < \varepsilon,$$

$$f(c-) = L \iff \forall \varepsilon > 0 \exists \delta > 0 \forall x : 0 < c - x < \delta \implies |f(x) - L| < \varepsilon.$$

f right-continuous at $c \iff f(c+) = f(c)$, f left-continuous at $c \iff f(c-) = f(c)$.

At an endpoint of a domain interval, continuity is interpreted relative to that interval and uses only the available side.

1. Calculus Review

One-sided limits and monotone functions (continued)

Theorem 1.79 (Sequential and two-sided criteria). [Ex. 1.43][Th. 1.18(iv)]

For f defined to the right of c ,

$$f(c+) = L \iff \forall (x_n) : x_n > c, x_n \rightarrow c \implies f(x_n) \rightarrow L.$$

Existence of a finite right-hand limit is also equivalent to finite convergence of $f(x_n)$ for every sequence approaching c from the right. The corresponding statements hold from the left. If f is defined on both sides, $\lim_{x \rightarrow c} f(x) = L \iff f(c-) = f(c+) = L$. If $f(c)$ is defined, then f continuous at $c \iff f(c-) = f(c) = f(c+)$.

Definition 1.80 (Finite one-sided discontinuities). Under the chapter's use of "jump discontinuity," $f(c-), f(c+) \in \mathbb{R}$, $f(c-) \neq f(c)$ or $f(c+) \neq f(c)$.

The following more specific distinctions are useful:

$$\begin{aligned} \text{removable discontinuity: } & f(c-) = f(c+) \neq f(c), \\ \text{nonzero jump: } & f(c-) \neq f(c+). \end{aligned}$$

The chapter's broad convention includes the removable case.

Definition 1.81 (Monotone functions). For $f : I \rightarrow \mathbb{R}$,

$$\begin{aligned} f \text{ increasing} & \iff \forall x, y \in I : x < y \implies f(x) \leq f(y), \\ f \text{ decreasing} & \iff \forall x, y \in I : x < y \implies f(x) \geq f(y). \end{aligned}$$

Strict monotonicity replaces the conclusion by a strict inequality.

Proposition 1.82 (One-sided limits of monotone functions). [Prop. 1.19][Ex. 1.49]

If $f : (a, b) \rightarrow \mathbb{R}$ is increasing and $c \in (a, b)$, then

$$\begin{aligned} f(c-) &= \sup\{f(t) : a < t < c\} \in \mathbb{R}, \\ f(c+) &= \inf\{f(t) : c < t < b\} \in \mathbb{R}, \\ f(c-) &\leq f(c) \leq f(c+). \end{aligned}$$

Consequently, f continuous at $c \iff f(c-) = f(c+)$,

and a discontinuity has $f(c-) < f(c+)$. For decreasing f , $f(c-) = \inf_{a < t < c} f(t)$, $f(c+) = \sup_{c < t < b} f(t)$, $f(c-) \geq f(c) \geq f(c+)$.

Thus a monotone function has finite one-sided limits at every interior point and only nonzero jumps at its interior discontinuities.

Proposition 1.83 (Limits at infinity). [Ex. 1.44]

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is increasing and bounded, then $\lim_{x \rightarrow +\infty} f(x) = \sup_{x \in \mathbb{R}} f(x)$, $\lim_{x \rightarrow -\infty} f(x) = \inf_{x \in \mathbb{R}} f(x)$.

For decreasing bounded f , these two extrema are interchanged. Without boundedness, the increasing-function formulas hold as extended limits. For a finite L , the limits at infinity mean

$$\begin{aligned} \lim_{x \rightarrow +\infty} f(x) = L & \iff \forall \varepsilon > 0 \exists M \in \mathbb{R} \forall x > M : |f(x) - L| < \varepsilon, \\ \lim_{x \rightarrow -\infty} f(x) = L & \iff \forall \varepsilon > 0 \exists M \in \mathbb{R} \forall x < M : |f(x) - L| < \varepsilon. \end{aligned}$$

Definition 1.84 (Infinite function limits). If f is defined on a punctured neighborhood of c ,

$$\begin{aligned} \lim_{x \rightarrow c} f(x) = +\infty & \iff \forall H \in \mathbb{R} \exists \delta > 0 : 0 < |x - c| < \delta \implies f(x) > H, \\ \lim_{x \rightarrow c} f(x) = -\infty & \iff \forall H \in \mathbb{R} \exists \delta > 0 : 0 < |x - c| < \delta \implies f(x) < H. \end{aligned}$$

Each implication is required for every x in the domain. One-sided infinite limits replace $0 < |x - c| < \delta$ by the corresponding one-sided inequality. At $+\infty$, replace it by $x > M$ for some $M \in \mathbb{R}$; at $-\infty$, replace it by $x < M$.

Lemma 1.85 (Symmetric continuity is weaker than continuity). [Ex. 1.42]

For f defined near c , $\lim_{h \rightarrow 0} |f(c+h) - f(c)| = 0 \implies \lim_{h \rightarrow 0} |f(c+h) - f(c-h)| = 0$.

The reverse implication fails. For $f = \mathbf{1}_{\{c\}}$, $f(c+h) = f(c-h) = 0$ ($h \neq 0$), $f(c) = 1$.

1. Calculus Review

Continuity consequences and important examples (continued)

Lemma 1.86 (Two continuous branches on rational and irrational points). [Ex. 1.47][Consequence]

Let $u, v \in C([0, 1])$, and set

$$F(x) = \begin{cases} u(x), & x \in \mathbb{Q}, \\ v(x), & x \notin \mathbb{Q}. \end{cases}$$

Then $\text{Cont}(F) = \{c \in [0, 1] : u(c) = v(c)\}$.

Example 1.87 (Standard continuity sets). [Ex. 1.47]

On $[0, 1]$, define

$$D(x) = \begin{cases} 0, & x \in \mathbb{Q}, \\ 1, & x \notin \mathbb{Q}, \end{cases} \quad G(x) = xD(x), \quad H(x) = \begin{cases} x, & x \in \mathbb{Q}, \\ 1-x, & x \notin \mathbb{Q}. \end{cases}$$

Then $\text{Cont}(D) = \emptyset$, $\text{Cont}(G) = \{0\}$, $\text{Cont}(H) = \{1/2\}$.

Replacing a rational branch by the irrational branch, and conversely, leaves these continuity sets unchanged.

Example 1.88 (Thomae's function). [Ex. 1.47]

For $x \in [0, 1]$, define

$$T(x) = \begin{cases} 1/q, & x = m/q \in \mathbb{Q}, \quad m \in \mathbb{Z}, \quad q \in \mathbb{N}, \quad \gcd(|m|, q) = 1, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Then $\text{Cont}(T) = [0, 1] \setminus \mathbb{Q}$.

In fact, $\lim_{x \rightarrow c, x \in [0, 1]} T(x) = 0$ for every $c \in [0, 1]$, with one-sided interpretation at 0 and 1.

Definition 1.89 (Injective, surjective, bijective). For $f : A \rightarrow B$,

$$\begin{aligned} f \text{ injective} &\iff \forall x, y \in A : f(x) = f(y) \implies x = y, \\ f \text{ surjective} &\iff \forall b \in B \exists a \in A : f(a) = b, \\ f \text{ bijective} &\iff f \text{ injective and surjective.} \end{aligned}$$

Every strictly monotone function is injective.

Theorem 1.90 (Increasing extension from rational points). [Ex. 1.50]

Let $D = [0, 1] \cap \mathbb{Q}$, and let $f : D \rightarrow \mathbb{R}$ be increasing. Set $g_-(x) = \sup\{f(q) : q \in D, q \leq x\}$, $g_+(x) = \inf\{f(q) : q \in D, x \leq q\}$.

These quantities are finite for every $x \in [0, 1]$, and $g_-(x) \leq g_+(x)$, g_-, g_+ are increasing, $g_-(q) = g_+(q) = f(q)$ ($q \in D$).

Thus both are increasing extensions. Every increasing extension g satisfies $g_-(x) \leq g(x) \leq g_+(x)$ ($x \in [0, 1]$).

In particular, the extension is unique if and only if $g_- = g_+$ on $[0, 1]$.

Proposition 1.91 (Right-continuous regularization). [Ex. 1.51]

Let $f : [a, b] \rightarrow \mathbb{R}$ be increasing, $a < b$, and define

$$g(x) = \begin{cases} f(x+) = \inf_{x < t \leq b} f(t), & a \leq x < b, \\ f(b), & x = b. \end{cases}$$

Then g is increasing, $g(x+) = g(x)$ ($a \leq x < b$), $f(x) \leq g(x)$.

At every point where f is continuous relative to $[a, b]$, $g(x) = f(x)$.

Corollary 1.92 (Left-continuous version). [Consequence]

For the same f , set

$$h(x) = \begin{cases} f(a), & x = a, \\ f(x-) = \sup_{a \leq t < x} f(t), & a < x \leq b. \end{cases}$$

Then h is increasing and left-continuous on $(a, b]$, and $h(x) \leq f(x) \leq g(x)$ ($x \in [a, b]$).

At interior continuity points, $h(x) = f(x) = g(x)$.

2. Countable and Uncountable Sets

Cardinality and countability

Definition 2.1 (Equivalence, cardinality, and function sets). For sets A, B ,

$$A \sim B \iff \exists f : A \rightarrow B \text{ bijective}, \quad |A| \leq |B| \iff \exists f : A \rightarrow B \text{ injective},$$

$$\mathcal{P}(A) = \{E : E \subset A\}, \quad B^A = \{f : f : A \rightarrow B\}, \quad |A| < |B| \iff |A| \leq |B| \wedge A \not\sim B.$$

A is finite iff $A \sim \{1, \dots, n\}$ for some $n \geq 0$, with $\{1, \dots, 0\} = \emptyset$. It is countable iff it admits an injection into $\mathbb{N}$, countably infinite iff $A \sim \mathbb{N}$, and uncountable iff it is not countable.

$$\aleph_0 = |\mathbb{N}|, \quad \mathfrak{c} = |\mathbb{R}|, \quad |A| + |B| = |(A \times \{0\}) \cup (B \times \{1\})|, \quad |A| |B| = |A \times B|.$$

Set-theoretic assertions use the usual axiom of choice.

Proposition 2.2 (No universal set). With all variables ranging over sets,

$$\neg \exists U \forall A : A \in U.$$

Lemma 2.3 (Subsets and images of countable sets). [Lemma 2.2][Ex. 2.7]

Every subset of a countable set is countable. If $A \subset \mathbb{N}$ is infinite, then $A = \{a_1 < a_2 < \dots\}$ and $A \sim \mathbb{N}$. The image of a countable set under any function is countable. If $g : C \rightarrow A$ is injective and A is countable, then C is countable.

Theorem 2.4 (Countable unions and finite products). [Th. 2.6; Cor. 2.7][Ex. 2.3]

$$(\forall n \in \mathbb{N}, A_n \text{ countable}) \implies \bigcup_{n \geq 1} A_n \text{ countable}.$$

Finite products of countable sets are countable. Consequently, $\mathbb{Z} \sim \mathbb{Q} \sim \mathbb{N}$, $\mathbb{N}^k \sim \mathbb{N}$ ($k \geq 1$), $\aleph_0^k = \aleph_0$, $\aleph_0 + \aleph_0 = \aleph_0$.

Proposition 2.5 (Infinite sets and countable modifications). [Ex. 2.2,4–6,14,17]

If A is infinite, then

$$\begin{aligned} \forall n \in \mathbb{N} \exists E \subset A : |E| = n, \quad \exists S \subset A : S \sim \mathbb{N}, \\ A \sim A \setminus \{a\} \quad (a \in A), \quad A \text{ infinite} \iff \exists B \subsetneq A : A \sim B. \end{aligned}$$

If B is countable, then $A \sim A \cup B$. If A is uncountable and B is countable, then $A \sim A \setminus B$, and $A \setminus B$ is uncountable. Every infinite A has a partition $A = \bigsqcup_{n \geq 1} A_n$ with every A_n infinite.

Theorem 2.6 (Uncountability and Cantor's theorem). [Th. 2.9; Cor. 2.10; Th. 2.12]

$$|\mathbb{R}| > \aleph_0, \quad |\mathbb{R} \setminus \mathbb{Q}| = \mathfrak{c}.$$

For every set A , no $F : A \rightarrow \mathcal{P}(A)$ is surjective; equivalently, $|A| < |\mathcal{P}(A)|$.

Theorem 2.7 (Cantor–Schröder–Bernstein). [Th. 2.13]

$$(\exists f : A \hookrightarrow B) \wedge (\exists g : B \hookrightarrow A) \implies A \sim B.$$

Corollary 2.8 (Continuum cardinalities). [Ex. 2.8–10,19]

For every nondegenerate real interval I , and every $k \in \mathbb{N}$, $I \sim \mathbb{R} \sim \mathbb{R}^k \sim \mathbb{R}^{\mathbb{N}} \sim \{0, 1\}^{\mathbb{N}} \sim \mathcal{P}(\mathbb{N})$,

$$\mathfrak{c} = 2^{\aleph_0}, \quad \mathfrak{c}^k = \mathfrak{c}, \quad \mathfrak{c}^{\aleph_0} = \mathfrak{c}, \quad \mathfrak{c}\aleph_0 = \mathfrak{c}.$$

For any A , the map $E \mapsto \mathbf{1}_E$ is a bijection $\mathcal{P}(A) \rightarrow \{0, 1\}^A$.

Proposition 2.9 (Countable exceptional families). [Ex. 2.12,15–16]

For every integer base $p \geq 2$, the real numbers with nonunique fractional base- p expansions form a countable set; even their union over all bases is countable. Eventually periodic expansions represent rational numbers, and their set is countable. Every pairwise disjoint family of nonempty open intervals in $\mathbb{R}$ is countable.

$$\mathcal{A} := \{z \in \mathbb{C} : \exists P \in \mathbb{Z}[t] \setminus \{0\}, P(z) = 0\} \text{ is countable.}$$

The real transcendental numbers have cardinality $\mathfrak{c}$.

2. Countable and Uncountable Sets

Cardinality and countability (continued)

Proposition 2.10 (Explicit transcendental numbers).

$$\left\{ \sum_{n \geq 1} a_n 10^{-n!} : a_n \in \{1, \dots, 9\} \right\} \subset \mathbb{R} \setminus \mathcal{A}, \quad e, \pi \in \mathbb{R} \setminus \mathcal{A}.$$

The displayed family has cardinality $\mathfrak{c}$.

Proposition 2.11 (Two uncountable constructions). [Ex. 2.18,20]

$$\left| \left\{ \sum_{n \geq 1} a_n 10^{-n} : a_n \in \{0, 1, 2, 4, 5, 6, 8, 9\} \right\} \right| = \mathfrak{c}.$$

There exists $\{S_\alpha : \alpha \in \{0, 1\}^{\mathbb{N}}\} \subset \mathcal{P}(\mathbb{N})$ such that each S_α is infinite and $\alpha \neq \beta \implies |S_\alpha \cap S_\beta| < \infty$.

Definition 2.12 (Ternary Cantor set). Put $C_0 = [0, 1]$ and, for $n \geq 1$,

$$C_n = \bigcup_{(a_1, \dots, a_n) \in \{0, 2\}^n} \left[\sum_{j=1}^n a_j 3^{-j}, \sum_{j=1}^n a_j 3^{-j} + 3^{-n} \right], \quad \mathcal{K} = \bigcap_{n \geq 0} C_n.$$

The n -th stage contains 2^n disjoint intervals of length 3^{-n} . The discarded gaps at stage n number 2^{n-1} , each of length 3^{-n} .

$$\sum_{n=1}^{\infty} 2^{n-1} 3^{-n} = 1, \quad \mathcal{K} = \frac{1}{3}\mathcal{K} \cup \left(\frac{2}{3} + \frac{1}{3}\mathcal{K}\right).$$

Theorem 2.13 (Digit description and cardinality). [Th. 2.15; Cor. 2.16]

$$x \in \mathcal{K} \iff x = \sum_{n \geq 1} 2b_n 3^{-n} \quad \text{for some } (b_n) \in \{0, 1\}^{\mathbb{N}}.$$

This representation using only digits 0, 2 is unique. The coding map $(b_n) \mapsto \sum 2b_n 3^{-n}$ is a bijection, and $|\mathcal{K}| = \mathfrak{c}$.

Proposition 2.14 (Topology, endpoints, and differences). [Ex. 2.21–25]

$\mathcal{K}$ is compact, perfect, nowhere dense, and has Lebesgue measure zero:

$$\mathcal{K}' = \mathcal{K}, \quad \text{int}_{\mathbb{R}} \mathcal{K} = \emptyset, \quad m(\mathcal{K}) = 0, \quad \mathcal{K} - \mathcal{K} = [-1, 1], \quad \mathcal{K} + \mathcal{K} = [0, 2].$$

Its endpoints form the countable dense subset $E = \mathcal{K} \cap \{a3^{-n} : a \in \mathbb{Z}, n \geq 0\}$.

Every point of $\mathcal{K}$ is a limit of distinct endpoints. Every discarded gap is

$$(s + 3^{-n}, s + 2 \cdot 3^{-n}), \quad s = \sum_{j=1}^{n-1} a_j 3^{-j}, \quad a_j \in \{0, 2\}.$$

A finite ternary expansion ending in two or more consecutive 1's is outside $\mathcal{K}$. Also $\text{Cont}(\mathbf{1}_{\mathcal{K}}) = \mathbb{R} \setminus \mathcal{K}$.

Definition 2.15 (Cantor function). [Cor. 2.16 and extension after Exercise 27]

Define $F_0 : \mathcal{K} \rightarrow [0, 1]$ by

$$F_0 \left(\sum_{n \geq 1} 2b_n 3^{-n} \right) = \sum_{n \geq 1} b_n 2^{-n}.$$

Its extension $F : [0, 1] \rightarrow [0, 1]$ is $F(x) = \sup\{F_0(t) : t \in \mathcal{K}, t \leq x\}$.

Theorem 2.16 (Properties of the Cantor function). [Ex. 2.26–29]

F is continuous, increasing, and surjective, with $F(0) = 0$, $F(1) = 1$. It is constant on the closure of every discarded gap. For $x < y$ in $\mathcal{K}$, $F_0(x) = F_0(y) \iff (x, y)$ is one discarded gap.

The lower endpoint x in this equality case has two ternary representations (the word “binary” in Exercise 2.26 is erroneous). If $x, y \in \mathcal{K}$ and $|x - y| < 3^{-n}$, then they lie in the same component of C_n , and $|F_0(x) - F_0(y)| \leq 2^{-n}$. For $x > 0$ in $\mathcal{K}$, $F_0(x) = \sup\{F_0(t) : t \in \mathcal{K}, t < x\}$.

At $x = 0$, use $F_0(0) = 0$ instead of the supremum of the empty set. Finally $F'(x) = 0$ for $x \in (0, 1) \setminus \mathcal{K}$, so $F' = 0$ Lebesgue-almost everywhere on $[0, 1]$.

2. Countable and Uncountable Sets

The Cantor set and the Cantor function (continued)

Proposition 2.17 (Generalized Cantor sets). [Ex. 2.30]

Fix $0 < \alpha < 1$. At stage n , remove the middle open interval of length $\alpha 3^{-n}$ from each of the 2^{n-1} surviving intervals. Let $\mathcal{K}_\alpha$ be the intersection of the surviving stages. Then

$$\text{length of each stage-}n \text{ interval} = \frac{1 - \alpha + \alpha(2/3)^n}{2^n},$$

$$\mathcal{K}_\alpha \text{ is compact, perfect, nowhere dense,} \quad |\mathcal{K}_\alpha| = \mathfrak{c}, \quad m(\mathcal{K}_\alpha) = 1 - \alpha.$$

Proposition 2.18 (Omitting one digit). [Ex. 2.31]

For an integer $p \geq 3$ and $d \in \{0, \dots, p-1\}$, put

$$K_{p,d} = \left\{ \sum_{n \geq 1} a_n p^{-n} : a_n \in \{0, \dots, p-1\} \setminus \{d\} \right\}.$$

Then $K_{p,d}$ is compact, $|K_{p,d}| = \mathfrak{c}$, and $m(K_{p,d}) = 0$. At stage n , it is covered by $(p-1)^n$ closed intervals of length p^{-n} . If a representation with integer part 1 is allowed at the endpoint 1, the digit-omission set may differ from this fractional coding only at that endpoint; the cardinality and measure conclusions are identical.

Theorem 2.19 (Discontinuity sets of monotone functions). [Th. 2.17][Ex. 2.32,34]

If $f : I \rightarrow \mathbb{R}$ is monotone on an interval I , its relative discontinuity set $D(f)$ is countable. Thus $\text{Cont}(f)$ is dense in I . At an interior discontinuity the finite one-sided limits differ; for increasing f , $f(x-) \leq f(x) \leq f(x+)$, $f(x-) < f(x+)$.

Conversely, every countable $D \subset \mathbb{R}$ is the discontinuity set of some increasing function $\mathbb{R} \rightarrow \mathbb{R}$, and even of a strictly increasing function. For distinct $(x_n)_{n \in J}$ enumerating D , with $J \subset \mathbb{N}$, choose $c_n > 0$ with $\sum_{n \in J} c_n < \infty$. One such increasing function is

$$h(x) = \sum_{n \in J, x_n \leq x} c_n, \quad h(x+) = h(x), \quad h(x_n) - h(x_n-) = c_n.$$

Then $D(h) = D$, and $x \mapsto x + h(x)$ is strictly increasing with the same discontinuity set.

Corollary 2.20 (Monotone surjections between closed intervals). [Cor. 2.18]

If $a \leq b$, $c \leq d$, and $f : [a, b] \rightarrow [c, d]$ is monotone and surjective, then f is continuous relative to $[a, b]$.

Proposition 2.21 (Total jump bound). [Ex. 2.33,35]

For increasing $f : [a, b] \rightarrow \mathbb{R}$, and any finite set of interior points S ,

$$\sum_{x \in S} (f(x+) - f(x-)) \leq f(b) - f(a).$$

The analogous inequality for decreasing f uses absolute values. Let $\{x_n\}$ enumerate the relative discontinuities, and set

$$a_n = \begin{cases} f(x_n) - f(x_n-), & x_n > a, \\ 0, & x_n = a, \end{cases} \quad b_n = \begin{cases} f(x_n+) - f(x_n), & x_n < b, \\ 0, & x_n = b. \end{cases}$$

Then $a_n, b_n \geq 0$, and

$$\sum_n (a_n + b_n) \leq f(b) - f(a).$$

Theorem 2.22 (Continuous part plus pure jumps). [Ex. 2.36]

With the preceding notation,

$$h(x) = \sum_{x_n \leq x} a_n + \sum_{x_n < x} b_n, \quad g(x) = f(x) - h(x).$$

Both h and g are increasing on $[a, b]$, g is continuous relative to $[a, b]$, and $f = g + h$.

3. Metrics and Norms

Metrics, pseudometrics, and constructions

Definition 3.1 (Metric and pseudometric). A metric on X is a function $d : X \times X \rightarrow [0, \infty)$ satisfying, for all $x, y, z \in X$, $d(x, y) = 0 \iff x = y$, $d(x, y) = d(y, x)$, $d(x, z) \leq d(x, y) + d(y, z)$.

A pseudometric satisfies $d(x, x) = 0$, symmetry, and the triangle inequality, but may have $d(x, y) = 0$ for $x \neq y$. All distances are finite. The metric subspace on $A \subset X$ uses $d|_{A \times A}$.

Example 3.2 (Elementary metrics). [Ex. 3.1]

$$d_{\text{disc}}(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y, \end{cases} \quad d_{\mathbb{R}}(x, y) = |x - y|, \quad d_{(0, \infty)}(x, y) = |x^{-1} - y^{-1}|.$$

Lemma 3.3 (Distance inequalities and minimal axioms). [Ex. 3.2–4]

$$|d(x, z) - d(y, z)| \leq d(x, y), \quad |d(x, y) - d(u, v)| \leq d(x, u) + d(y, v).$$

If $d : X^2 \rightarrow \mathbb{R}$ satisfies $d(x, y) = 0 \iff x = y$ and $d(x, y) \leq d(x, z) + d(y, z)$, then d is a metric. If a nonnegative symmetric separating function satisfies the reversed triangle inequality $d(x, y) \geq d(x, z) + d(z, y)$, then $|X| \leq 1$.

Proposition 3.4 (Metric transforms). [Ex. 3.5–7]

If $\phi : [0, \infty) \rightarrow [0, \infty)$ is increasing, $\phi(0) = 0$, $\phi(t) > 0$ for $t > 0$, and $\phi(s + t) \leq \phi(s) + \phi(t)$, then $\phi \circ d$ is a metric. Sufficient conditions for the last inequality include a decreasing ratio $\phi(t)/t$, or concavity with $\phi(0) = 0$. In particular,

$$d^\theta \quad (0 < \theta \leq 1), \quad \frac{d}{1 + d}, \quad \min\{d, 1\}$$

are metrics equivalent to d .

Proposition 3.5 (Combining metrics). [Ex. 3.8]

If d_1, d_2 are metrics on X , then $d_1 + d_2$ and $\max\{d_1, d_2\}$ are metrics; $\min\{d_1, d_2\}$ need not be. On $\{a, b, c\}$, take the edge lengths $(d_1(a, b), d_1(b, c), d_1(a, c)) = (1, 3, 3)$, $(d_2(a, b), d_2(b, c), d_2(a, c)) = (3, 1, 3)$.

Their minimum has lengths $(1, 1, 3)$. Also $|x - y|^2$ fails the triangle inequality on $\mathbb{R}$.

Definition 3.6 (Hilbert cube and countable products). [Ex. 3.9–11]

$$H = [-1, 1]^{\mathbb{N}}, \quad d_H(x, y) = \sum_{n \geq 1} 2^{-n} |x_n - y_n|,$$

$$d_\Pi(x, y) = \sum_{n \geq 1} 2^{-n} \frac{|x_n - y_n|}{1 + |x_n - y_n|} \quad (x, y \in \mathbb{R}^{\mathbb{N}}).$$

Both are metrics; d_H also restricts to a metric on $\{0, 1\}^{\mathbb{N}}$. For $x, y \in H$ and $M_k = \max_{1 \leq j \leq k} |x_j - y_j|$, $2^{-k} M_k \leq d_H(x, y) \leq (1 - 2^{-k}) M_k + 2^{1-k} \leq M_k + 2^{1-k}$.

The tail term 2^{1-k} is required for the cube $[-1, 1]^{\mathbb{N}}$. For either product metric, convergence is equivalent to coordinatewise convergence.

Example 3.7 (Metrics on continuous functions). [Ex. 3.12–13]

For $a < b$, on $C([a, b])$,

$$d_\infty(f, g) = \max_{t \in [a, b]} |f(t) - g(t)|, \quad d_1(f, g) = \int_a^b |f - g| dt,$$

are metrics. On $C([0, 1])$, another metric is $\int_0^1 \min\{|f(t) - g(t)|, 1\} dt$.

Definition 3.8 (Norm and seminorm). Let V be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. A norm $\|\cdot\| : V \rightarrow [0, \infty)$ satisfies $\|x\| = 0 \iff x = 0$, $\|\lambda x\| = |\lambda| \|x\|$, $\|x + y\| \leq \|x\| + \|y\|$.

A seminorm satisfies the latter two axioms and $\|0\| = 0$, with no separation requirement. The textbook also calls it a pseudonorm. $d(x, y) = \|x - y\|$ is the induced metric; every linear subspace inherits the norm.

Proposition 3.9 (Norm-induced metrics). [Ex. 3.16]

A metric d on V is induced by a norm iff $d(x, y) = d(x - y, 0)$, $d(\lambda x, \lambda y) = |\lambda| d(x, y)$.

Then $\|x\| = d(x, 0)$. The discrete metric on a nonzero real vector space is not induced by a norm.

3. Metrics and Norms

Normed spaces and classical inequalities (continued)

Definition 3.10 (Finite and infinite sequence norms). [Ex. 3.17,20–23]

For $1 \leq p < \infty$,

$$\|x\|_p = \left(\sum_{j=1}^k |x_j|^p \right)^{1/p} \quad (x \in \mathbb{F}^k), \quad \|x\|_\infty = \max_j |x_j|,$$

$$\ell^p = \left\{ x \in \mathbb{F}^{\mathbb{N}} : \sum_{n \geq 1} |x_n|^p < \infty \right\}, \quad \|x\|_p = \left(\sum_{n \geq 1} |x_n|^p \right)^{1/p},$$

$$\ell^\infty = \{x : \sup_n |x_n| < \infty\}, \quad \|x\|_\infty = \sup_n |x_n|, \quad c_0 = \{x : x_n \rightarrow 0\}, \quad c_{00} = \{x : x_n = 0 \text{ eventually}\}.$$

These are normed vector spaces (c_0, c_{00} use $\|\cdot\|_\infty$). For $A = (a_{ij}) \in \mathbb{F}^{k \times m}$, the following is a norm:

$$\|A\| = \max_{1 \leq i \leq k} \left(\sum_{j=1}^m |a_{ij}|^2 \right)^{1/2}.$$

Proposition 3.11 (Norm comparisons). [Ex. 3.18,22–23,44]

For $x \in \mathbb{F}^k$,

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1, \quad \|x\|_1 \leq k\|x\|_\infty, \quad \|x\|_2 \leq \sqrt{k}\|x\|_\infty, \quad \|x\|_1 \leq \sqrt{k}\|x\|_2.$$

More generally, $1 \leq p < q \leq \infty$ implies $\|x\|_q \leq \|x\|_p \leq k^{1/p-1/q}\|x\|_q$ ($x \in \mathbb{F}^k$),

with $1/\infty = 0$. For infinite sequences,

$$\ell^p \subsetneq \ell^q \subsetneq c_0 \subsetneq \ell^\infty \quad (1 \leq p < q < \infty), \quad \|x\|_q \leq \|x\|_p \quad (x \in \ell^p).$$

Lemma 3.12 (Cauchy–Schwarz). [Lemma 3.3][Ex. 3.19]

For $x, y \in \ell^2$,

$$\sum_n |x_n y_n| \leq \|x\|_2 \|y\|_2, \quad |\langle x, y \rangle| \leq \|x\|_2 \|y\|_2, \quad \langle x, y \rangle = \sum_n x_n \overline{y_n}.$$

Equality in the first inequality iff the nonnegative sequences $(|x_n|), (|y_n|)$ are proportional, including either zero sequence. Equality in the second iff x, y are linearly dependent. For real sequences, $\sum x_n y_n = \|x\|_2 \|y\|_2$ iff one is a nonnegative multiple of the other (including zero cases).

Lemma 3.13 (Power-sum estimate). [Lemma 3.5]

For $p \geq 1$ and $a, b \geq 0$, $(a+b)^p \leq 2^{p-1}(a^p + b^p) \leq 2^p(a^p + b^p)$.

In particular, $x, y \in \ell^p \implies x+y \in \ell^p$.

Lemma 3.14 (Young). [Lemma 3.6]

Let $1 < p < \infty$, $q = p/(p-1)$. For $a, b \geq 0$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}, \quad \text{equality} \iff a^p = b^q \iff b = a^{p-1}.$$

Lemma 3.15 (Hölder). [Lemma 3.7][Ex. 3.24]

For conjugate exponents $1 \leq p, q \leq \infty$, $1/p + 1/q = 1$,

$$\sum_n |x_n y_n| \leq \|x\|_p \|y\|_q \quad (x \in \ell^p, y \in \ell^q).$$

If $1 < p, q < \infty$ and both sequences are nonzero, equality iff $|x_n|^p = \lambda |y_n|^q$ for all n , for some $\lambda > 0$.

Theorem 3.16 (Minkowski). [Ths. 3.4,3.8]

For $1 \leq p \leq \infty$ and $x, y \in \ell^p$, $\|x+y\|_p \leq \|x\|_p + \|y\|_p$.

Thus every ℓ^p is a normed vector space.

Proposition 3.17 (Integral analogues). [Ex. 3.25]

For $a < b$, $f, g \in C([a, b])$, and $1 \leq p < \infty$, put

$$\|f\|_p = \left(\int_a^b |f(t)|^p dt \right)^{1/p}, \quad \|f\|_\infty = \max_{[a,b]} |f|.$$

These are norms. For conjugate p, q , including $(1, \infty)$,

$$\int_a^b |fg| dt \leq \|f\|_p \|g\|_q, \quad \|f+g\|_p \leq \|f\|_p + \|g\|_p.$$

Proposition 3.18 (Limits of power sums). [Ex. 3.26]

For $a, b > 0$ and $L_p = (a^p + b^p)^{1/p}$,

$$\lim_{p \rightarrow \infty} L_p = \max\{a, b\}, \quad \lim_{p \rightarrow -\infty} L_p = \min\{a, b\}, \quad \lim_{p \rightarrow -1} L_p = \frac{ab}{a+b},$$

$$\lim_{p \rightarrow 0+} L_p = +\infty, \quad \lim_{p \rightarrow 0-} L_p = 0.$$

3. Metrics and Norms

Balls, boundedness, convergence, and Cauchy sequences (continued)

Definition 3.19 (Balls, neighborhoods, diameter). [Ex. 3.14–15, 27–30]

$$B_r^X(x) = \{y \in X : d(x, y) < r\}, \quad \overline{B}_r^X(x) = \{y \in X : d(x, y) \leq r\} \quad (r > 0).$$

U is a neighborhood of x iff $B_r(x) \subset U$ for some $r > 0$. $\text{diam } A = \sup\{d(a, b) : a, b \in A\}$, $\text{diam } \emptyset = 0$.

For nonempty A ,

$$A \text{ bounded} \iff \exists x \in X \exists r > 0 : A \subset B_r(x) \iff \text{diam } A < \infty.$$

Then $\sup_{a \in A} d(x, a) < \infty$ for every $x \in X$. Finite unions of bounded sets are bounded. Also $\text{diam } B_r(x) \leq 2r$, $A \subset B \implies \text{diam } A \leq \text{diam } B$.

If $A \neq \emptyset$ and $\text{diam } A < r$, then $A \subset B_r(a)$ for every $a \in A$.

Lemma 3.20 (Diameter of unions). [Ex. 3.31]

$$A \cap B \neq \emptyset \implies \text{diam}(A \cup B) \leq \text{diam } A + \text{diam } B.$$

Without the intersection hypothesis, take $A = \{0\}$, $B = \{1\}$ in $\mathbb{R}$.

Proposition 3.21 (Normed-space balls). [Ex. 3.32]

$$B_r(x) = x + B_r(0), \quad B_r(0) = rB_1(0).$$

In $\mathbb{R}$, $B_r(x) = (x - r, x + r)$. In a discrete space, $B_r(x) = \{x\}$ for $0 < r \leq 1$, and $B_r(x) = X$ for $r > 1$.

Definition 3.22 (Metric convergence and Cauchy condition).

$$x_n \rightarrow x \iff d(x_n, x) \rightarrow 0 \iff \forall \varepsilon > 0 \exists N \forall n \geq N : d(x_n, x) < \varepsilon.$$

Equivalently, (x_n) is eventually in every neighborhood of x .

$$(x_n) \text{ Cauchy} \iff \forall \varepsilon > 0 \exists N \forall m, n \geq N : d(x_m, x_n) < \varepsilon \iff \text{diam}\{x_n : n \geq N\} \rightarrow 0.$$

Proposition 3.23 (Basic sequence facts). [Ex. 3.33–37, 39]

Limits are unique. Subsequences of a convergent sequence have the same limit. Convergent sequences are Cauchy; Cauchy sequences are bounded. $x_n \rightarrow x$, $y_n \rightarrow y \implies d(x_n, y_n) \rightarrow d(x, y)$.

A Cauchy sequence with a subsequence converging to x converges to x . Also $x_n \rightarrow x$ iff every subsequence has a further subsequence converging to x .

Lemma 3.24 (Rapidly Cauchy subsequences). [Ex. 3.38]

A sequence has a Cauchy subsequence iff it has a subsequence (x_{n_k}) with $d(x_{n_k}, x_{n_{k+1}}) < 2^{-k}$ for all k . More generally, $\sum_k d(y_k, y_{k+1}) < \infty$ implies that (y_k) is Cauchy.

Proposition 3.25 (Coordinate convergence and truncations). [Ex. 3.40–41]

On $\mathbb{F}^k$, convergence and the Cauchy condition are equivalent to their coordinatewise versions. In ℓ^p , norm convergence implies coordinatewise convergence; the converse fails for the unit vectors e_n , since $(e_n)_j \rightarrow 0$ (j fixed), $\|e_n\|_p = 1$.

For $P_N x = (x_1, \dots, x_N, 0, \dots)$,

$$\|x - P_N x\|_p^p = \sum_{n > N} |x_n|^p \rightarrow 0 \quad (x \in \ell^p, p < \infty),$$

$$\|x - P_N x\|_\infty = \sup_{n > N} |x_n| \rightarrow 0 \iff x \in c_0.$$

If $x^{(n)} \rightarrow x$, $y^{(n)} \rightarrow y$ in ℓ^2 , then $\langle x^{(n)}, y^{(n)} \rangle \rightarrow \langle x, y \rangle$.

Example 3.26 (Failures outside complete or compact spaces). In $(0, 1]$, $(1/n)$ is Cauchy but has no limit in the space. In $\mathbb{R}$ with the discrete metric, (n) is bounded but has no Cauchy subsequence. In any discrete space, convergence and the Cauchy condition each mean eventual constancy.

Definition 3.27 (Equivalent metrics and norms). [Ex. 3.42–45]

Metrics d, ρ on X are equivalent iff

$$\forall (x_n) \subset X \forall x \in X : d(x_n, x) \rightarrow 0 \iff \rho(x_n, x) \rightarrow 0.$$

Two norms are equivalent iff their induced metrics are equivalent, or equivalently iff they have the same sequences converging to 0. The usual and discrete metrics on $\mathbb{N}$ are equivalent; every metric on a finite set is equivalent to the discrete metric.

Proposition 3.28 (Finite product metrics). [Ex. 3.46]

For $(X, d), (Y, \rho)$,

$$\begin{aligned} D_1((x, y), (u, v)) &= d(x, u) + \rho(y, v), \\ D_2((x, y), (u, v)) &= \sqrt{d(x, u)^2 + \rho(y, v)^2}, \\ D_\infty((x, y), (u, v)) &= \max\{d(x, u), \rho(y, v)\}. \end{aligned}$$

These are equivalent metrics, with $D_\infty \leq D_2 \leq D_1 \leq 2D_\infty$. Convergence and the Cauchy condition are each equivalent to the corresponding conditions in both coordinates. The same holds for any finite product.

4. Open Sets and Closed Sets

Open and closed sets

Definition 4.1 (Open, closed, and clopen). For $A \subset (X, d)$,

$$A \text{ open} \iff \forall x \in A \exists r > 0 : B_r(x) \subset A, \quad A \text{ closed} \iff X \setminus A \text{ open}.$$

A set is clopen iff it is both open and closed. Write $\tau_d = \{U \subset X : U \text{ open}\}$.

Theorem 4.2 (Operations on open and closed sets). [Ex. 4.1–2]

$\emptyset, X$, and every open ball are open. Arbitrary unions and finite intersections of open sets are open. Arbitrary intersections and finite unions of closed sets are closed. Every finite set is closed. If F is closed and G open, $F \setminus G$ is closed and $G \setminus F$ is open. Finite products of open sets are open in the product, and finite products of closed sets are closed. In a discrete space, every subset is clopen.

Example 4.3 (Infinite operations). $\bigcap_{n \geq 1} (-1/n, 1/n) = \{0\}$ is not open, $\bigcup_{n \geq 1} [1/n, 1] = (0, 1]$ is not closed.

The set $(0, 1] \subset \mathbb{R}$ is neither open nor closed.

Theorem 4.4 (Open subsets of the real line). [Th. 4.6][Ex. 4.7–8, 25]

Every open $U \subset \mathbb{R}$ is a countable disjoint union of maximal open intervals, possibly unbounded. This family of maximal intervals is unique. Also $U = \bigcup \{(r, s) : r, s \in \mathbb{Q}, r < s, (r, s) \subset U\}$, $|\tau_{\mathbb{R}}| = \mathfrak{c}$.

Every open real set is a countable union of closed bounded intervals. Every closed real interval is a countable intersection of open intervals. The only clopen subsets of $\mathbb{R}$ are $\emptyset$ and $\mathbb{R}$.

Theorem 4.5 (Sequential characterizations). [Ths. 4.7, 4.9]

$$U \text{ open} \iff [x_n \rightarrow x \in U \implies \exists N \forall n \geq N : x_n \in U].$$

The following are equivalent:

$$F \text{ closed}; \quad [\forall r > 0, B_r(x) \cap F \neq \emptyset \implies x \in F]; \quad [(x_n) \subset F, x_n \rightarrow x \in X \implies x \in F].$$

Corollary 4.6 (Equivalent metrics). [Ex. 4.3]

$d \sim \rho \iff \tau_d = \tau_\rho \iff d, \rho$ have the same closed sets.

Proposition 4.7 (Intersections of open sets). [Ex. 4.4, 29]

For any $A \subset X$, $A = \bigcap_{x \in X \setminus A} (X \setminus \{x\})$.

Every closed set in a metric space is a countable intersection of open sets; every open set is a countable union of closed sets.

Definition 4.8 (Interior and closure).

$$\begin{aligned} \text{int}_X A &= \bigcup \{U \subset A : U \in \tau_d\} = \{x \in A : \exists r > 0, B_r(x) \subset A\}, \\ \overline{A}^X &= \text{cl}_X A = \bigcap \{F \supset A : F \text{ closed in } X\}. \end{aligned}$$

When the ambient space is unambiguous, write A° and $\overline{A}$.

Theorem 4.9 (Closure tests). [Prop. 4.10; Cor. 4.11]

$$x \in \overline{A} \iff \forall r > 0 : B_r(x) \cap A \neq \emptyset \iff \exists (a_n) \subset A : a_n \rightarrow x.$$

4. Open Sets and Closed Sets

Interior, closure, boundary, and derived sets (continued)

Proposition 4.10 (Closure and interior identities). [Ex. 4.17,20–24]

$$\begin{aligned} A^\circ &\subset A \subset \overline{A}, \quad \overline{\overline{A}} = \overline{A}, \quad (A^\circ)^\circ = A^\circ, \\ A \text{ open} &\iff A = A^\circ, \quad A \text{ closed} \iff A = \overline{A}, \\ \overline{A} &= X \setminus \text{int}(X \setminus A), \quad A^\circ = X \setminus \overline{X \setminus A}, \\ \overline{A \cup B} &= \overline{A} \cup \overline{B}, \quad \overline{A \cap B} \subset \overline{A} \cap \overline{B}, \\ \text{int}(A \cap B) &= A^\circ \cap B^\circ, \quad A^\circ \cup B^\circ \subset \text{int}(A \cup B). \end{aligned}$$

Both operations preserve inclusion. The inclusion for intersections of closures may be strict: $A = (0, 1)$, $B = (1, 2)$. The inclusion for interiors of unions may be strict: $A = \mathbb{Q}$, $B = \mathbb{R} \setminus \mathbb{Q}$. $\overline{A} \subset \overline{B}$ need not imply $A \subset B$, e.g. $A = \mathbb{R}$, $B = \mathbb{Q}$.

Proposition 4.11 (Closed balls, extrema, and diameter). [Ex. 4.15–19,42]

$\overline{B}_r(x) = \{y : d(x, y) \leq r\}$ is closed, and $\overline{B}_r(x) \subset \overline{B}_r(x)$. Equality holds in every normed vector space for $r > 0$; it may fail in a discrete space at $r = 1$. $\text{diam } \overline{A} = \text{diam } A$.

For nonempty bounded $E \subset \mathbb{R}$, $\inf E, \sup E \in \overline{E}$; if E is closed, both are in E .

Definition 4.12 (Distance to a set and between sets). [Ex. 4.26–28,32]

For nonempty $A, B \subset X$, $d(x, A) = \inf_{a \in A} d(x, a)$, $d(A, B) = \inf_{a \in A, b \in B} d(a, b)$.

$$d(x, A) = 0 \iff x \in \overline{A}, \quad |d(x, A) - d(y, A)| \leq d(x, y), \quad d(x, A) = d(x, \overline{A}).$$

Thus $\{d(\cdot, A) < \varepsilon\}$ is open and $\{d(\cdot, A) \leq \varepsilon\}$ closed. They contain A whenever $\varepsilon > 0$. Disjoint closed sets can have zero distance: in $\mathbb{R}^2$, $A = \{(t, 0) : t \geq 1\}$, $B = \{(t, 1/t) : t \geq 1\}$.

Proposition 4.13 (Countable descriptions and locally finite unions). [Ex. 4.29–30]

For nonempty closed F , $F = \bigcap_{n \geq 1} \{x : d(x, F) < 1/n\}$.

For open U with nonempty complement, $U = \bigcup_{n \geq 1} \{x : d(x, X \setminus U) \geq 1/n\}$.

A locally finite family of closed sets has closed union. In particular, if $F_n \subset (n, n+1)$ is closed in $\mathbb{R}$ for every $n \in \mathbb{Z}$, $\bigcup_{n \in \mathbb{Z}} F_n$ is closed. Countable disjoint closed unions need not be closed: $\bigcup_{n \geq 1} \{1/n\}$.

Proposition 4.14 (Metric separation). [Ex. 4.23,31]

If $x \neq y$, there exist open U, V with $x \in U$, $y \in V$, and $\overline{U} \cap \overline{V} = \emptyset$. If F is closed and $x \notin F$, the same holds with $F \subset V$. Any two disjoint closed sets have open neighborhoods with disjoint closures.

Definition 4.15 (Limit points and isolated points). [Ex. 4.33–35,40]

$$A' = \{x \in X : \forall r > 0, (B_r(x) \setminus \{x\}) \cap A \neq \emptyset\}.$$

For $x \in A$, x is isolated in A iff $B_r(x) \cap A = \{x\}$ for some $r > 0$, equivalently $x \notin A'$.

$$x \in A' \iff \forall r > 0, |B_r(x) \cap A| = \infty \iff \exists \text{ pairwise distinct } a_n \in A \setminus \{x\} : a_n \rightarrow x.$$

A' is closed, $\overline{A} = A \cup A'$, and A closed $\iff A' \subset A$.

Proposition 4.16 (Accumulation and perfect sets). [Ex. 4.36–40]

If $x_n \rightarrow x$, then $\{x\} \cup \{x_n : n \geq 1\}$ is closed (and compact). Every bounded infinite subset of $\mathbb{R}$ has a limit point. A set P is perfect iff P is closed and $P = P'$, including $P = \emptyset$. Every nonempty perfect subset of $\mathbb{R}$ is uncountable. Every real set has at most countably many isolated points; hence every uncountable real set has a limit point. $\mathcal{K}$ is perfect, as is $\mathbb{R} \times \{0\} \subset \mathbb{R}^2$.

Definition 4.17 (Boundary). [Ex. 4.41–45]

$$\partial_X A = \{x : \forall r > 0, B_r(x) \cap A \neq \emptyset \text{ and } B_r(x) \cap A^c \neq \emptyset\}.$$

$$\partial A = \overline{A} \cap \overline{A^c} = \overline{A} \setminus A^\circ = \partial(A^c),$$

$$\overline{A} = A^\circ \sqcup \partial A, \quad X = A^\circ \sqcup \partial A \sqcup (A^c)^\circ,$$

$$\partial A \text{ closed, } A \text{ closed} \iff \partial A \subset A, \quad \partial A = \emptyset \iff A \text{ clopen.}$$

For nonempty bounded $E \subset \mathbb{R}$, $\inf E, \sup E \in \partial E$. Examples: $\partial \mathbb{R} = \emptyset$, $\partial \mathbb{Q} = \mathbb{R}$.

4. Open Sets and Closed Sets

Density, separability, bases, and nowhere dense sets (continued)

Definition 4.18 (Dense sets and separability). [Ex. 4.46,48]

$$\begin{aligned} D \text{ dense in } X &\iff \overline{D} = X \iff \forall x \in X \forall r > 0 : B_r(x) \cap D \neq \emptyset \\ &\iff \forall U \in \tau_d \setminus \{\emptyset\} : U \cap D \neq \emptyset \iff \text{int}(D^c) = \emptyset \iff \forall x \in X \exists (d_n) \subset D : d_n \rightarrow x. \end{aligned}$$

X is separable iff it has a countable dense subset.

Proposition 4.19 (Density identities). [Ex. 4.47,66–67]

If G is open and D dense, then $\overline{G \cap D}^X = \overline{G}^X$, and $G \cap D$ is dense in G . For arbitrary G , this may fail: $G = \{\sqrt{2}\}$, $D = \mathbb{Q}$ in $\mathbb{R}$. If $A \subset B \subset X$, A is dense in B , and B dense in X , then A is dense in X .

Proposition 4.20 (Standard separable spaces). [Ex. 4.48–50,53,68]

$\mathbb{R}^k, H, \ell^p$ ($1 \leq p < \infty$), c_0 , and $C([a, b])$ with the supremum norm are separable. Countable dense sets include $\mathbb{Q}^k \subset \mathbb{R}^k$, $\{(q_1, \dots, q_N, 0, \dots) : N \in \mathbb{N}, q_j \in \mathbb{Q}\}$

in real ℓ^p, c_0 (and with $q_j \in [-1, 1]$ in H). In $C([0, 1])$, continuous piecewise linear functions with rational breakpoints and rational vertex values form a countable dense family. ℓ^∞ is not separable: its subset $\{0, 1\}^{\mathbb{N}}$ is uncountable and has pairwise distances 1. The closure of every separable subset is separable.

Definition 4.21 (Open base and second countability). [Ex. 4.69]

$\mathcal{B} \subset \tau_d$ is an open base iff

$$\forall U \in \tau_d \forall x \in U \exists B \in \mathcal{B} : x \in B \subset U.$$

A space is second countable iff it has a countable open base. For metric spaces, X separable $\iff X$ second countable.

If D is countable and dense, $\{B_r(x) : x \in D, r \in \mathbb{Q}, r > 0\}$ is a base.

Corollary 4.22 (Countability consequences). [Ex. 4.51–52][Consequence]

A separable metric space has at most countably many isolated points; every family of pairwise disjoint nonempty open sets is countable. Every metric subspace of a separable metric space is separable.

Definition 4.23 (Nowhere density). [Ex. 4.54,57,60]

For $A \subset X$, the following are equivalent:

$$\text{int } \overline{A} = \emptyset; \quad X \setminus \overline{A} \text{ dense}; \quad \overline{A} \text{ contains no nonempty open set};$$

every nonempty open U contains a nonempty open $V \subset U$ with $V \cap A = \emptyset$; every such U contains an open ball disjoint from A . These conditions define “ A nowhere dense in X .” For closed A , they reduce to $A^\circ = \emptyset$.

Proposition 4.24 (Nowhere dense examples and operations). [Ex. 4.54–59]

$\{x\}$ is nowhere dense in X iff x is not isolated in X . Finite unions and arbitrary subsets of nowhere dense sets are nowhere dense. Every finite real set and the Cantor set are nowhere dense; $\mathbb{Q}$ is not. If A is closed, ∂A is nowhere dense. The boundary of an arbitrary set need not be nowhere dense, since $\partial \mathbb{Q} = \mathbb{R}$. For an enumeration (q_n) of $\mathbb{Q}$, $U = \bigcup_{n \geq 1} (q_n - 2^{-n}, q_n + 2^{-n})$ is a proper open dense subset of $\mathbb{R}$, and U^c is closed and nowhere dense.

Theorem 4.25 (Subspace formulas). [Prop. 4.13][Ex. 4.61–64]

$$\text{For } E \subset A \subset X \text{ and } x \in A, \quad B_r^A(x) = A \cap B_r^X(x), \quad \overline{E}^A = A \cap \overline{E}^X.$$

$$G \text{ open in } A \iff G = A \cap U \quad (U \text{ open in } X),$$

$$F \text{ closed in } A \iff F = A \cap C \quad (C \text{ closed in } X).$$

If A is open in X , relative openness of $G \subset A$ is equivalent to ambient openness. If A is closed, the analogous equivalence holds for closed sets. In general $\text{int}_A E \neq A \cap \text{int}_X E$: take $E = A = \mathbb{Q} \subset \mathbb{R}$.

Lemma 4.26 (Extending disjoint relative open sets). [Ex. 4.65]

If $G, H \subset A \subset X$ are disjoint and open in A , there exist disjoint open $U, V \subset X$ such that $G = A \cap U$, $H = A \cap V$.

5. Continuity

Continuity and its characterizations

Definition 5.1 (Continuity at a point). [Ex. 5.9–10]

For $f : (X, d) \rightarrow (Y, \rho)$ and $x \in X$,

$$f \text{ continuous at } x \iff \forall \varepsilon > 0 \exists \delta > 0 \forall u \in X : d(u, x) < \delta \implies \rho(f(u), f(x)) < \varepsilon.$$

Equivalently, $f(B_\delta(x)) \subset B_\varepsilon(f(x))$, or $B_\delta(x) \subset f^{-1}(B_\varepsilon(f(x)))$, for a suitable δ . Continuity of $f|_A$ at $x \in A$ uses only $u \in A$. Every function is continuous at every isolated domain point.

Theorem 5.2 (Equivalent continuity criteria). [Th. 5.1][Ex. 5.30]

For $f : X \rightarrow Y$, the following are equivalent:

- (i) f is continuous at every point of X .
- (ii) $x_n \rightarrow x$ implies $f(x_n) \rightarrow f(x)$.
- (iii) Every inverse image of an open set is open.
- (iv) Every inverse image of a closed set is closed.
- (v) $f(\overline{A}) \subset \overline{f(A)}$ for every $A \subset X$.
- (vi) $f^{-1}(B^\circ) \subset (f^{-1}(B))^\circ$ for every $B \subset Y$.

Proposition 5.3 (Images and inverse images). [Ex. 5.1]

For $f : S \rightarrow T$, $A, B \subset S$, $C, D \subset T$,

$$\begin{aligned} A &\subset f^{-1}(f(A)), & f(f^{-1}(C)) &\subset C, \\ f\left(\bigcup_i A_i\right) &= \bigcup_i f(A_i), & f\left(\bigcap_i A_i\right) &\subset \bigcap_i f(A_i), \\ f^{-1}\left(\bigcup_i C_i\right) &= \bigcup_i f^{-1}(C_i), & f^{-1}\left(\bigcap_i C_i\right) &= \bigcap_i f^{-1}(C_i), \\ f(A) \setminus f(B) &\subset f(A \setminus B), & f^{-1}(C \setminus D) &= f^{-1}(C) \setminus f^{-1}(D), \\ f^{-1}(T \setminus C) &= S \setminus f^{-1}(C). \end{aligned}$$

Equality $A = f^{-1}(f(A))$ for every A , or equality for all binary intersection images, is equivalent to injectivity. Equality $f(f^{-1}(C)) = C$ for every C is equivalent to surjectivity. The arbitrary intersection image identity holds for injective f when the index set is nonempty; for the empty family it also requires surjectivity.

Proposition 5.4 (Characteristic functions). [Ex. 5.2–5]

$$\begin{aligned} \mathbf{1}_{A \cup B} &= \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B, & \mathbf{1}_{A \cap B} &= \mathbf{1}_A \mathbf{1}_B, & \mathbf{1}_{A \setminus B} &= \mathbf{1}_A(1 - \mathbf{1}_B), \\ \mathbf{1}_C \circ f &= \mathbf{1}_{f^{-1}(C)}, & D(\mathbf{1}_A) &= \partial A, & \text{Cont}(\mathbf{1}_A) &= A^\circ \cup (A^c)^\circ. \end{aligned}$$

$\mathbf{1}_A$ is continuous iff A is clopen. On $\mathbb{R}$, the only continuous characteristic functions are the constants 0, 1.

Proposition 5.5 (Real-valued continuity tests). [Ex. 5.6–8, 14, 17]

$f : X \rightarrow \mathbb{R}$ is continuous iff $\{f > a\}$ and $\{f < a\}$ are open for every $a \in \mathbb{R}$; it suffices to use $a \in \mathbb{Q}$. If $f(x) > 0$ and f is continuous at x , then $f > 0$ on some neighborhood of x . If continuous $f, g : X \rightarrow Y$ agree on a dense subset, they agree on X . If continuous real f is nonnegative on a dense subset, then $f \geq 0$ everywhere. Strict positivity does not extend in the same way: $f(t) = (t - \sqrt{2})^2$. $|C(\mathbb{R})| = \mathfrak{c}$.

Lemma 5.6 (Composition and products). [Lemmas 5.7–5.8]

If $f : X \rightarrow Y$ is continuous at x and $g : Y \rightarrow Z$ continuous at $f(x)$, then $g \circ f$ is continuous at x . For finitely many maps $f_j : X \rightarrow Y_j$, $x \mapsto (f_1(x), \dots, f_k(x))$ continuous $\iff$ each f_j continuous.

Proposition 5.7 (Restriction and pasting). [Ex. 5.11, 31]

Restrictions of continuous functions are continuous. If $X = \bigcup_{i \in I} U_i$ with every U_i open and every $f|_{U_i}$ continuous, then f is continuous. The same conclusion holds for a finite closed cover, or a locally finite closed cover. An arbitrary countable closed cover does not suffice: on $X = \{0\} \cup \{1/n : n \geq 1\}$, set $f(0) = 0$, $f(1/n) = 1$. Continuity of $f|_A, f|_B$ alone does not imply continuity on $A \cup B$.

5. Continuity

Continuity and its characterizations (continued)

Proposition 5.8 (Additive functions and graphs). [Ex. 5.15–16]

If $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x+y) = f(x) + f(y)$ and is continuous at one point, then $f(x) = cx$ for all x , with $c = f(1)$. For arbitrary $f : \mathbb{R} \rightarrow \mathbb{R}$, the following are equivalent:

$$f \text{ continuous}; \quad x \mapsto (x, f(x)) \text{ continuous}; \quad \{(x, y) : y \leq f(x)\} \text{ and } \{(x, y) : y \geq f(x)\} \text{ closed.}$$

Continuous f has closed graph. The converse fails for $f(0) = 0$, $f(x) = 1/x$ for $x \neq 0$.

Proposition 5.9 (Continuous images and density). [Ex. 5.17–18]

If D is dense in X and $f : X \rightarrow Y$ continuous, then $f(D)$ is dense in $f(X)$. Continuous images of separable spaces are separable.

Example 5.10 (A map from irrationals onto rationals). [Ex. 5.12]

Let $I = [0, 1] \setminus \mathbb{Q}$, $r_n = 1 - 1/n$, and let (q_n) enumerate $[0, 1] \cap \mathbb{Q}$. The map $g(x) = q_n$ ($x \in I \cap (r_n, r_{n+1})$) is continuous on I and maps it onto $[0, 1] \cap \mathbb{Q}$.

Definition 5.11 (Lipschitz maps and isometries). [Ex. 5.19,25]

$f : (X, d) \rightarrow (Y, \rho)$ is L -Lipschitz iff $\rho(f(x), f(y)) \leq Ld(x, y)$ ($x, y \in X$), $0 \leq L < \infty$.

It is an isometry into Y iff equality holds with $L = 1$; it is an isometry onto iff it is also surjective. Lipschitz maps are uniformly continuous; isometries are injective and are homeomorphisms onto their images.

Theorem 5.12 (Distance functions and zero sets). [Props. 5.3–5.4][Ex. 5.35–36,39,53]

For nonempty $A \subset X$, $x \mapsto d(x, A)$ is 1-Lipschitz and its zero set is $\bar{A}$. A subset $F \subset X$ is closed iff $F = u^{-1}(\{0\})$ for some continuous $u : X \rightarrow \mathbb{R}$; for $F = \emptyset$ use $u = 1$. A set U is open iff $U = \{u > 0\}$ for some continuous nonnegative u , taking $u = d(\cdot, U^c)$ when $U^c \neq \emptyset$. $x_n \rightarrow x \iff u(x_n) \rightarrow u(x)$ for every $u \in C(X)$.

Proposition 5.13 (Continuous separation of closed sets). [Ex. 5.21,37–38,40]

If $E, F \subset X$ are nonempty, closed, and disjoint, then

$$u(x) = \frac{d(x, E)}{d(x, E) + d(x, F)}$$

is continuous, $0 \leq u \leq 1$, $u^{-1}(\{0\}) = E$, $u^{-1}(\{1\}) = F$. The sets $\{u < 1/3\}$, $\{u > 2/3\}$ are open neighborhoods of E, F with disjoint closures. This conclusion does not require $d(E, F) > 0$.

Theorem 5.14 (Extensions). [Ex. 5.41–42]

If $C \subset \mathbb{R}$ is closed and $f : C \rightarrow \mathbb{R}$ continuous, there exists a continuous $g : \mathbb{R} \rightarrow \mathbb{R}$ with $g|_C = f$. On a bounded complementary interval (a, b) with $a, b \in C$, one may set

$$g(x) = \frac{b-x}{b-a}f(a) + \frac{x-a}{b-a}f(b).$$

On unbounded complementary rays use the adjacent endpoint value. Every L -Lipschitz $f : \mathbb{Q} \rightarrow \mathbb{R}$ has a unique L -Lipschitz extension to $\mathbb{R}$, given by $g(x) = \lim_n f(q_n)$ whenever $q_n \rightarrow x$.

Definition 5.15 (Semicontinuity). [Ex. 5.32–33]

For real $f : X \rightarrow \mathbb{R}$,

$$\begin{aligned} f \text{ lower semicontinuous} &\iff \{f \leq a\} \text{ closed for all } a \in \mathbb{R} \\ &\iff f(x) \leq \liminf_n f(x_n) \text{ whenever } x_n \rightarrow x, \\ f \text{ upper semicontinuous} &\iff \{f \geq a\} \text{ closed for all } a \in \mathbb{R} \\ &\iff f(x) \geq \limsup_n f(x_n) \text{ whenever } x_n \rightarrow x. \end{aligned}$$

Continuity is equivalent to both conditions. A finite-valued pointwise supremum of continuous functions is lower semicontinuous; a finite-valued infimum is upper semicontinuous.

Definition 5.16 (Homeomorphism and open/closed maps). [Ex. 5.56]

A bijection $f : X \rightarrow Y$ is a homeomorphism iff f and f^{-1} are continuous. An embedding is a homeomorphism onto its image with the subspace metric. An open map sends every open subset of X to an open subset of Y ; a closed map sends every closed subset of X to a closed subset of Y .

5. Continuity

Homeomorphisms, embeddings, and topological properties (continued)

Theorem 5.17 (Homeomorphism criteria). [Th. 5.5][Ex. 5.43–44,52,54,57–58]

For a bijection $f : X \rightarrow Y$, the following are equivalent:

- (i) f is a homeomorphism.
- (ii) $x_n \rightarrow x \iff f(x_n) \rightarrow f(x)$.
- (iii) G is open in X iff $f(G)$ is open in Y .
- (iv) F is closed in X iff $f(F)$ is closed in Y .
- (v) $\rho(f(x), f(y))$ defines a metric equivalent to d .
- (vi) $f(\bar{A}) = \overline{f(A)}$ for every $A \subset X$.
- (vii) $u \in C(Y) \iff u \circ f \in C(X)$ for every $u : Y \rightarrow \mathbb{R}$.

For an arbitrary bijection, openness, closedness, and continuity of f^{-1} are equivalent. Homeomorphism is an equivalence relation.

Proposition 5.18 (Basic classifications). [Ex. 5.45–46,48,55,60–61]

Every map from a discrete space is continuous. Every continuous map from $\mathbb{R}$ to a discrete space is constant. $\mathbb{N} \cong \{1/n : n \geq 1\} \cong \{e_n : n \geq 1\}$, $\mathbb{R} \cong (0, 1) \cong (0, \infty)$,

where $\cong$ denotes homeomorphism. $\mathbb{R}$ is isometric to neither of the latter two intervals. Any two nondegenerate bounded intervals of the same endpoint type are homeomorphic; the types $[a, b]$, $(a, b]$, and (a, b) , with $a < b$, are pairwise nonhomeomorphic. Every metric space is homeomorphic to a bounded metric space, using $d/(1+d)$. Separability is invariant under homeomorphisms. $X \cong (X, d_{\text{disc}}) \iff \tau_d = \mathcal{P}(X) \iff$ every $u : X \rightarrow \mathbb{R}$ is continuous.

Example 5.19 (Homeomorphic spaces with inequivalent metrics). [Ex. 5.50]

On $X = \{0\} \cup \{1/n : n \geq 1\}$, let d be usual and let T interchange 0, 1 while fixing all other points. Set $\rho(x, y) = d(Tx, Ty)$. Then $T : (X, \rho) \rightarrow (X, d)$ is an isometry, but d, ρ are inequivalent: $1/n \rightarrow 0$ in d , while $1/n \rightarrow 1$ in ρ .

Theorem 5.20 (Separable metric spaces embed into the Hilbert cube). [Ex. 5.51]

Let (X, ρ) be separable with $\rho \leq 1$, and let (x_n) be dense. $\Phi(x) = (\rho(x, x_n))_{n \geq 1} \in H$, $d_H(\Phi(x), \Phi(y)) \leq \rho(x, y)$. Φ is injective and is a homeomorphism onto $\Phi(X)$. Thus every separable metric space embeds into H .

Example 5.21 (Continuity, openness, and closedness). [Ex. 5.56,59]

$x \mapsto x^2 : \mathbb{R} \rightarrow \mathbb{R}$ is continuous but not open; $x \mapsto \arctan x : \mathbb{R} \rightarrow \mathbb{R}$ is continuous but not closed. The identity from usual $\mathbb{R}$ to discrete $\mathbb{R}$ is both open and closed but not continuous. The projection $\mathbb{R}^2 \rightarrow \mathbb{R}$, $(x, y) \mapsto x$, is continuous and open but not closed. The onto map $x \mapsto x^2 : [-1, 2] \rightarrow [0, 4]$ is continuous and closed but not open.

Theorem 5.22 (Pointwise algebra and lattice operations). $C(X) = \{f : X \rightarrow \mathbb{R} : f \text{ continuous}\}$ is a real vector space, a commutative unital algebra, and a vector lattice under pointwise operations. For $f, g \in C(X)$, $af + bg$, fg , $|f|$, $\max\{f, g\}$, $\min\{f, g\} \in C(X)$.

If $g \neq 0$ everywhere, $f/g \in C(X)$. Also $\max\{f, g\} = \frac{1}{2}(f + g + |f - g|)$, $\min\{f, g\} = \frac{1}{2}(f + g - |f - g|)$.

The maps $(s, t) \mapsto s + t, st, \max\{s, t\}, \min\{s, t\}$ are continuous.

Proposition 5.23 (Supremum norm structure). [Ex. 5.62–63]

On $C([a, b])$, $a < b$, $\|fg\|_\infty \leq \|f\|_\infty \|g\|_\infty$, $\|\max\{f, g\}\|_\infty \leq \max\{\|f\|_\infty, \|g\|_\infty\}$, $|f| \leq |g| \implies \|f\|_\infty \leq \|g\|_\infty$.

For $\sigma(t) = a + t(b - a)$, the map $T : C([a, b]) \rightarrow C([0, 1])$, $Tf = f \circ \sigma$, is an onto linear isometry, an algebra isomorphism, and a lattice isomorphism. For finite $X = \{x_1, \dots, x_k\}$, $f \mapsto (f(x_1), \dots, f(x_k))$ identifies $C(X)$ with $\mathbb{R}^k$, preserving pointwise algebra, order, and the supremum norm.

Theorem 5.24 (Multiplicative linear functionals on $C(\mathbb{R})$). If $\varphi : C(\mathbb{R}) \rightarrow \mathbb{R}$ is real-linear and $\varphi(fg) = \varphi(f)\varphi(g)$ for all f, g , then either $\varphi = 0$, or

$$\exists! a \in \mathbb{R} \forall f \in C(\mathbb{R}) : \varphi(f) = f(a).$$

In the latter case $\varphi(1) = 1$. No continuity assumption on φ is needed.

Definition 5.25 (A metric on $C(\mathbb{R})$). [Ex. 5.64]

$$p_n(f - g) = \max_{|t| \leq n} |f(t) - g(t)|, \quad D(f, g) = \sum_{n \geq 1} 2^{-n} \frac{p_n(f - g)}{1 + p_n(f - g)}.$$

Each $p_n(f - g)$ is a pseudometric; D is a metric. $D(f_j, f) \rightarrow 0$ iff $f_j \rightarrow f$ uniformly on every bounded closed interval, equivalently on every compact subset of $\mathbb{R}$.

6. Connectedness

Connected sets and components

Definition 6.1 (Connectedness and separation). [Ex. 6.3–4]

$E \subset X$ is disconnected iff $E = A \sqcup B$, $A, B \neq \emptyset$, A, B open in E .

Equivalently, A, B may be required to be closed in E , or $E = A \sqcup B$, $A, B \neq \emptyset$, $A \cap \overline{B}^X = \emptyset = \overline{A}^X \cap B$.

E is connected iff no such separation exists, equivalently its only relatively clopen subsets are $\emptyset, E$. If disjoint open $U, V \subset X$ cover a connected E , then $E \subset U$ or $E \subset V$.

Lemma 6.2 (Ambient and relative separations). [Lemma 6.3][Ex. 6.1]

Disjoint relatively open subsets $G, H \subset E \subset X$ can be written $G = E \cap U, H = E \cap V$ for disjoint ambient open U, V . Thus connectedness of a metric subspace is independent of the ambient metric space containing it isometrically.

Theorem 6.3 (Connected real sets). For $E \subset \mathbb{R}$,

$$E \text{ connected} \iff \forall x, y \in E : [\min(x, y), \max(x, y)] \subset E.$$

Thus the nonempty connected real sets are precisely the intervals, including singletons. The empty set is connected. A discrete space is connected iff it has at most one point.

Definition 6.4 (Components and total disconnectedness). [Ex. 6.2]

For $x \in E$, its connected component is $C_E(x) = \bigcup \{A \subset E : x \in A, A \text{ connected}\}$.

Components are maximal connected subsets, are closed in E , and form a unique disjoint partition of E . For open subsets of $\mathbb{R}$, components are the maximal open intervals. E is totally disconnected iff every nonempty connected subset is a singleton. The Cantor set, $\mathbb{Q}$, and every discrete space are totally disconnected.

Proposition 6.5 (Unions and closure). [Ex. 6.5–10]

A, B connected, $A \cap B \neq \emptyset \implies A \cup B$ connected.

If a family of connected sets has a common point, its union is connected. If every pair of points of X belongs to some connected subset, X is connected. A connected, $A \subset B \subset \overline{A} \implies B$ connected.

In particular $\overline{A}$ is connected. If $E, F \neq \emptyset$ and $E \cup F$ is connected, then they are not separated, and $\overline{E} \cap \overline{F} \neq \emptyset$. Connected endpoints of inclusions do not suffice: $\{0\} \subset \{0, 1\} \subset [0, 1]$.

Lemma 6.6 (Continuous tests for connectedness). [Lemma 6.5][Ex. 6.11–12]

The following are equivalent:

- (i) X is connected.
- (ii) Every continuous $X \rightarrow \{0, 1\}$ is constant.
- (iii) Every continuous map from X into any discrete space is constant.
- (iv) If continuous $u : X \rightarrow \mathbb{R}$ never vanishes, then $u > 0$ throughout X or $u < 0$ throughout X , when $X \neq \emptyset$.

A connected metric space with at least two points has cardinality at least $\mathfrak{c}$, and in particular is uncountable.

Theorem 6.7 (Continuous images and the intermediate value property). [Th. 6.6; Cor. 6.7]

Continuous images of connected sets are connected. Connectedness and total disconnectedness are invariant under homeomorphisms. If $I \subset \mathbb{R}$ is an interval and $f : I \rightarrow \mathbb{R}$ continuous, then $f(I)$ is an interval. In particular, for $a < b$ in I and t between $f(a), f(b)$,

$$\exists c \in [a, b] : f(c) = t; \quad t \text{ strictly intermediate} \implies c \in (a, b).$$

Lemma 6.8 (Products and closed-set decompositions). [Lemma 6.9][Ex. 6.18]

Finite products of connected spaces are connected. If A, B are closed in X , and $A \cup B, A \cap B$ are connected, then A, B are connected.

6. Connectedness

Consequences for real functions (continued)

Proposition 6.9 (Fixed points and equal values). [Ex. 6.13–14]

Every continuous $f : [a, b] \rightarrow [a, b]$ has a fixed point. If $f : [0, 2] \rightarrow \mathbb{R}$ is continuous and $f(0) = f(2)$, then $f(x) = f(x+1)$ for some $x \in [0, 1]$.

Proposition 6.10 (Injectivity, openness, and monotonicity). [Ex. 6.15–16]

A continuous injective real function on an interval is strictly monotone, and is a homeomorphism onto its image. A continuous open map $f : \mathbb{R} \rightarrow \mathbb{R}$ is strictly monotone.

Proposition 6.11 (Obstructions to continuous maps). [Ex. 6.17, 19]

There is no continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying simultaneously $f(\mathbb{Q}) \subset \mathbb{R} \setminus \mathbb{Q}$, $f(\mathbb{R} \setminus \mathbb{Q}) \subset \mathbb{Q}$.

No continuous map $[0, 1] \rightarrow \mathbb{Q} \cap [0, 1]$ is onto, although such an onto continuous map exists on $[0, 1] \setminus \mathbb{Q}$.

Proposition 6.12 (Small fibers). [Ex. 6.20–21]

If continuous $f : [a, b] \rightarrow \mathbb{R}$ satisfies $|f^{-1}(\{y\})| \leq 2$ for every y , then some attained value has exactly one preimage, and f is piecewise strictly monotone on a finite partition. If $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $|f^{-1}(\{y\})| \in \{0, 2\}$ for every y , then $D(f)$ is infinite.

Definition 6.13 (Darboux property). [Ex. 6.22]

$f : I \rightarrow \mathbb{R}$, for an interval I , has the intermediate value (Darboux) property iff for every $a < b$ in I and $\min\{f(a), f(b)\} < t < \max\{f(a), f(b)\}$, $f(c) = t$ for some $c \in (a, b)$.

Theorem 6.14 (Darboux property and closed graph). [Ex. 6.22–23]

If $f : \mathbb{R} \rightarrow \mathbb{R}$ has the Darboux property and its graph is closed in $\mathbb{R}^2$, then f is continuous. If $f : I \rightarrow \mathbb{R}$ is differentiable on an open interval, then f' has the Darboux property, whether or not f' is continuous.

Definition 6.15 (Paths and path connectedness). [Ex. 6.27–28]

A path in X is a continuous $\gamma : [0, 1] \rightarrow X$. A path that is injective is an arc: it is a homeomorphism of $[0, 1]$ onto its image. Arcwise connectedness requires an arc between each pair of distinct points. X is path connected iff every $x, y \in X$ admit a path with $\gamma(0) = x$, $\gamma(1) = y$. Path connectedness implies connectedness. In a normed space, $\gamma(t) = (1-t)x + ty$ is a path, and an embedding if $x \neq y$. Every convex subset of a normed space is path connected; every normed space is connected.

Example 6.16 (Connected graphs and failure of path connectedness). [Ex. 6.25–26]

A continuous function can have connected range and disconnected domain: the constant map on $\{0, 1\}$. If the graph $\{(x, f(x)) : x \in D\}$ is connected, then D is connected. For continuous $f : D \rightarrow \mathbb{R}$, the graph is connected iff D is connected. The function $f(0) = 0$, $f(x) = \sin(1/x)$ for $0 < x \leq 1$ is discontinuous, but its graph is connected and is not path connected. Its closure is the closed connected set $\{(x, \sin(1/x)) : 0 < x \leq 1\} \cup (\{0\} \times [-1, 1])$.

Proposition 6.17 (Nonhomeomorphism results). [Ex. 6.24]

For $a < b$, the intervals $[a, b]$, $(a, b]$, (a, b) are pairwise nonhomeomorphic. $[0, 1]$ is not homeomorphic to $[0, 1]^2$, and $\mathbb{R}$ is not homeomorphic to $\mathbb{R}^k$ for $k \geq 2$. More generally $\mathbb{R}^k \not\cong \mathbb{R}^m$ for $k \neq m$. The connected open square $(0, 1)^2$ cannot be expressed as a disjoint union of Euclidean open balls.

Theorem 6.18 (Cantor surjections and space-filling curves). For every $k \geq 1$, there exists a continuous surjection $\mathcal{K} \rightarrow [0, 1]^k$, and a continuous surjection $[0, 1] \rightarrow [0, 1]^k$. For $t = \sum_{n \geq 1} 2b_n 3^{-n} \in \mathcal{K}$, one explicit map is

$$\Psi_k(t) = \left(\sum_{n \geq 1} b_{k(n-1)+j} 2^{-n} \right)_{j=1}^k.$$

Extending its coordinates linearly across every complementary gap gives an onto continuous path on $[0, 1]$. For $k \geq 2$, no continuous injective $[0, 1] \rightarrow [0, 1]^k$ can have dense image.

7. Completeness

Total boundedness

Definition 7.1 (Total boundedness and finite nets). [Ex. 7.4]

$A \subset X$ is totally bounded iff

$$\forall \varepsilon > 0 \exists \text{ finite } F \subset X : A \subset \bigcup_{x \in F} B_\varepsilon(x).$$

Such an F is an ε -net. For nonempty A , one may require $F \subset A$. Equivalently, for every $\varepsilon > 0$, A admits a finite cover by subsets of diameter $< \varepsilon$; it also admits such a cover by ambient closed sets of diameter $\leq \varepsilon$.

Proposition 7.2 (Elementary total boundedness facts). [Ex. 7.1–5,9–11]

Totally bounded sets are bounded. Subsets and finite unions of totally bounded sets are totally bounded; A is totally bounded iff $\overline{A}$ is. In $\mathbb{R}^k$, total boundedness is equivalent to boundedness. In a discrete metric space, total boundedness is equivalent to finiteness. Every totally bounded metric space is separable; the Hilbert cube H is totally bounded. Total boundedness is not invariant under homeomorphisms: $(0, 1) \cong \mathbb{R}$. The closed bounded set $\{e_n : n \geq 1\} \subset \ell^\infty$ is not totally bounded.

Theorem 7.3 (Sequential characterization). [Lemma 7.3; Th. 7.5][Ex. 7.6,8]

A totally bounded $\iff$ every sequence in A has a Cauchy subsequence.

Equivalently every sequence has a subsequence (x_{n_k}) with $d(x_{n_k}, x_{n_{k+1}}) < 2^{-k}$. The range of every Cauchy sequence is totally bounded. If A is not totally bounded, then for some $\varepsilon > 0$ it contains distinct a_n with $d(a_n, a_m) \geq \varepsilon$ for $n \neq m$. This set is a closed discrete subspace of X .

Corollary 7.4 (Bolzano–Weierstrass). [Cor. 7.6][Ex. 7.7]

Every bounded infinite subset of $\mathbb{R}$ has a limit point in $\mathbb{R}$.

Example 7.5 (Subsequence conclusions are optimal). The totally bounded range $\{-1, 1\}$ of $((-1)^n)$ does not make the full sequence Cauchy. The bounded unit-vector sequence in ℓ^1 has no Cauchy subsequence.

Definition 7.6 (Complete space and Banach space). X complete $\iff$ every Cauchy sequence in X converges in X .

A complete normed vector space is a Banach space. The empty metric space is complete; fixed-point assertions below explicitly require nonempty spaces.

Proposition 7.7 (Standard complete spaces). [Ex. 7.16–19,24,44]

$\mathbb{R}^k$, every discrete metric space, every ℓ^p for $1 \leq p \leq \infty$, c_0 , H , and $C([a, b])$ with the supremum norm are complete. For any set S , $B(S) = \{f : S \rightarrow \mathbb{R} : \sup_{s \in S} |f(s)| < \infty\}$, $\|f\|_\infty = \sup_{s \in S} |f(s)|$, is a Banach space (use $\|f\| = 0$ if $S = \emptyset$). The notation $B(S)$ replaces the textbook's $\ell^\infty(S)$.

Definition 7.8 (Pointwise and uniform convergence). [Consequence]

For maps $f_n, f : S \rightarrow (Y, \rho)$,

$$f_n \rightarrow f \text{ pointwise} \iff \forall s \in S : \rho(f_n(s), f(s)) \rightarrow 0,$$

$$f_n \rightarrow f \text{ uniformly} \iff \forall \varepsilon > 0 \exists N \forall n \geq N \forall s \in S : \rho(f_n(s), f(s)) < \varepsilon.$$

For bounded scalar-valued maps this is equivalent to $\|f_n - f\|_\infty \rightarrow 0$.

Lemma 7.9 (Uniform limits and bounded continuous functions). [Consequence]

A uniform limit of continuous maps $X \rightarrow Y$ is continuous. If Y is complete, every uniformly Cauchy sequence of maps $S \rightarrow Y$ has a uniform limit:

$$\forall \varepsilon > 0 \exists N \forall m, n \geq N \forall s \in S : \rho(f_m(s), f_n(s)) < \varepsilon.$$

Consequently $C_b(X, \mathbb{R}) = C(X, \mathbb{R}) \cap B(X)$ is a closed subspace of $B(X)$, and hence a Banach space under the supremum norm.

7. Completeness

Complete spaces and completeness criteria (continued)

Theorem 7.10 (Complete subspaces). [Th. 7.9][Ex. 7.12,17,35]

Every complete metric subspace $A \subset X$ is closed in X . If X is complete, then A complete $\iff A$ closed in X .

A finite product of nonempty metric spaces is complete iff every factor is complete. A normed space is complete iff its closed unit ball is complete.

Example 7.11 (Completeness and coordinate limits). [Ex. 7.13–15,33]

Completeness is not a topological invariant, nor is it preserved by arbitrary continuous images. Usual $\mathbb{R}$ is complete, whereas the equivalent metric $\rho(s, t) = |\arctan s - \arctan t|$ is incomplete. The metric $\sigma(s, t) = |s^3 - t^3|$ is complete. On $\mathbb{N}$, $d(m, n) = |1/m - 1/n|$ is equivalent to the usual metric but incomplete. c_{00} with the supremum norm is incomplete, and its completion is c_0 . The sequence $x^{(n)} = (1, \dots, 1, 0, \dots)$, with n ones, converges coordinatewise to $(1, 1, \dots)$, but is not Cauchy in c_0 or ℓ^∞ .

Lemma 7.12 (Cauchy comparisons and dense testing). [Ex. 7.20–23,28–29,34]

If $(x_n), (y_n)$ are Cauchy, then $(d(x_n, y_n))$ is Cauchy in $\mathbb{R}$. In a complete space, their limits agree iff $d(x_n, y_n) \rightarrow 0$. For dense $D \subset X$, X is complete iff every Cauchy sequence in D has a limit in X . Also the following characterize completeness:

- (i) Every sequence with $d(x_n, x_{n+1}) < 2^{-n}$ converges in X .
- (ii) Every countable closed subset of X is complete.
- (iii) Every closed ball in X is complete.

Theorem 7.13 (Cantor nested-set criterion). [Th. 7.11]

For a metric space X , the following are equivalent:

- (i) X is complete.
- (ii) Every decreasing sequence of nonempty closed sets $F_n \subset X$ with $\text{diam } F_n \rightarrow 0$ has intersection consisting of exactly one point.
- (iii) Every infinite totally bounded subset of X has a limit point in X .

If $\bigcap F_n = \{x\}$ in (ii), any choices $x_n \in F_n$ satisfy $x_n \rightarrow x$.

Example 7.14 (Nested-set hypotheses). [Ex. 7.26–27]

In complete $\mathbb{R}$, the decreasing nonempty sets $(0, 1/n)$ have diameters tending to zero but empty intersection. Closedness is essential. The closed sets $[n, \infty)$ have empty intersection. Even closed bounded sets do not suffice in a general complete space without shrinking diameter: $F_n = \{e_k : k \geq n\} \subset \ell^2$. In $\mathbb{R}$, nested nonempty closed bounded intervals do have nonempty intersection without shrinking diameter, but the intersection need not be a singleton. Boundedness alone cannot replace total boundedness in (iii), as $\{e_n\} \subset \ell^2$ has no limit point.

Proposition 7.15 (Open subsets admit complete equivalent metrics). [Ex. 7.30]

If (X, d) is complete, $G \subset X$ is open, and $F = X \setminus G \neq \emptyset$, then

$$\rho(x, y) = d(x, y) + \left| \frac{1}{d(x, F)} - \frac{1}{d(y, F)} \right| \quad (x, y \in G)$$

is a complete metric equivalent to $d|_G$. If $F = \emptyset$, use d .

Definition 7.16 (Vector-valued series). [Ex. 7.31]

$$\sum_{n \geq 1} x_n = x \text{ in } V \iff \left\| x - \sum_{n=1}^N x_n \right\| \rightarrow 0.$$

Absolute summability means $\sum_n \|x_n\| < \infty$. A convergent series has $\|x_n\| \rightarrow 0$. In a Banach space the Cauchy criterion is

$$\sum x_n \text{ converges} \iff \forall \varepsilon > 0 \exists N \forall m \geq n \geq N : \left\| \sum_{k=n}^m x_k \right\| < \varepsilon.$$

Theorem 7.17 (Banach's series criterion). [Th. 7.12][Ex. 7.32]

A normed space V is complete iff every absolutely summable series converges in V . In that case,

$$\left\| \sum_{n \geq 1} x_n \right\| \leq \sum_{n \geq 1} \|x_n\|, \quad \left\| \sum_{n > N} x_n \right\| \leq \sum_{n > N} \|x_n\|.$$

7. Completeness

Contractions, iteration, and differential equations (continued)

Definition 7.18 (Contraction and iterates). $T : X \rightarrow X$ is a strict contraction iff it is q -Lipschitz for some $0 \leq q < 1$. Define $T^0 = \text{id}$, $T^{n+1} = T \circ T^n$. The orbit of x_0 is $(T^n x_0)_{n \geq 0}$. A fixed point is $p \in X$ with $Tp = p$.

Theorem 7.19 (Banach contraction principle and error bounds). [Th. 7.13; equation 7.1 and following estimate]

Let $X \neq \emptyset$ be complete and $T : X \rightarrow X$ a q -contraction, $0 < q < 1$. There exists a unique fixed point p , and every orbit $x_n = T^n x_0$ converges to p . For $m > n \geq 0$,

$$\begin{aligned} d(x_m, x_n) &\leq \frac{q^n(1 - q^{m-n})}{1 - q} d(x_1, x_0), \\ d(x_n, p) &\leq \frac{q^n}{1 - q} d(x_1, x_0), \quad d(x_n, p) \leq \frac{q}{1 - q} d(x_n, x_{n-1}) \quad (n \geq 1), \\ d(x, p) &\leq \frac{d(x, Tx)}{1 - q}. \end{aligned}$$

For $q = 0$, T is constant and the fixed point is reached after one iteration.

Corollary 7.20 (Contractive iterates). [Ex. 7.41–42]

If $X \neq \emptyset$ is complete and some T^k is a strict contraction, then T has a unique fixed point, and all its orbits converge to that point. No separate continuity assumption on T is necessary. On $C([0, 1])$,

$$(Tf)(x) = \int_0^x f(t) dt, \quad \|T\| = 1, \quad \|T^n\| \leq 1/n!,$$

so T^2 is a strict contraction, and its unique fixed point is 0.

Example 7.21 (Necessary fixed-point hypotheses). [Ex. 7.43]

On incomplete $(0, 1)$, $T(x) = x/2$ is a strict contraction without fixed point. On complete $[1, \infty)$, $T(x) = x + 1/x$ satisfies $|T(x) - T(y)| < |x - y|$ for $x \neq y$, but has no fixed point. Thus pointwise strict distance decrease is weaker than a uniform contraction.

Definition 7.22 (Attracting and repelling fixed points). [Ex. 7.36–38]

A fixed point p is attracting if every sufficiently close initial point has orbit converging to p . A sufficient repelling condition is

$$\exists \delta > 0 \exists \lambda > 1 : \quad 0 < |x - p| < \delta \implies |T(x) - p| \geq \lambda |x - p|.$$

For a real map differentiable at p , $|T'(p)| < 1 \implies p$ attracting, $|T'(p)| > 1 \implies p$ repelling.

The value $|T'(p)| = 1$ is inconclusive: $\arctan x$ attracts to 0, $x + x^3$ repels from 0, and $x^2 + 1/4$ attracts to $1/2$ from the left and repels on the right. For $T(x) = x^2$, $T^n x = x^{2^n}$ for $n \geq 1$; $|x| < 1$ gives limit 0, $|x| > 1$ gives $+\infty$, and $x = \pm 1$ gives the eventual value 1.

Proposition 7.23 (Derivative criterion and numerical roots). [Ex. 7.39–40]

If continuous $T : [a, b] \rightarrow [a, b]$ is differentiable inside and $|T'(x)| \leq q < 1$, then T has a unique fixed point and iteration converges. If continuous $G : [a, b] \rightarrow \mathbb{R}$ is differentiable inside with $G(a) < 0 < G(b)$, $0 < m \leq G'(x) \leq M < \infty$, then it has a unique zero p . For $0 < \lambda \leq 1/M$, $T(x) = x - \lambda G(x)$ maps $[a, b]$ into itself and is a contraction with factor $1 - \lambda m$ (the zero-factor case is constant). Examples are

$$x_{n+1} = x_n - \frac{x_n^3 - 5}{8} \quad (x_0 \in [1, 2]), \quad x_n \rightarrow \sqrt[3]{5},$$

and

$$x_{n+1} = (x_n + 1)^{1/3} \quad (x_0 \in [1, 2]), \quad x_n \rightarrow \sqrt[3]{\frac{9 + \sqrt{69}}{18}} + \sqrt[3]{\frac{9 - \sqrt{69}}{18}},$$

the unique real root of $x^3 - x - 1 = 0$.

7. Completeness

Contractions, iteration, and differential equations (continued)

Theorem 7.24 (Picard–Lindelöf formulation). Suppose F is continuous on a rectangle around $(0, y_0)$ and uniformly Lipschitz in its second variable there. Then, for some $\delta > 0$, the initial value problem $y'(t) = F(t, y(t))$, $y(0) = y_0$ has a unique local solution within that rectangle. It is the fixed point of

$$(\mathcal{T}\phi)(t) = y_0 + \int_0^t F(s, \phi(s)) ds.$$

If F is continuous on $\mathbb{R}^2$ and globally L -Lipschitz in its second variable, then on $C([0, \delta])$, $\|\mathcal{T}\phi - \mathcal{T}\psi\|_\infty \leq L\delta\|\phi - \psi\|_\infty$, so $L\delta < 1$ gives a contraction. For $F(t, y) = 2t(1 + y)$, $y_0 = 0$,

$$\phi_0 = 0, \quad \phi_n(t) = \sum_{j=1}^n \frac{t^{2j}}{j!}, \quad y(t) = e^{t^2} - 1.$$

Definition 7.25 (Completion). A completion of (X, d) is a complete metric space $(\widehat{X}, \widehat{d})$ together with an isometric embedding $i : X \rightarrow \widehat{X}$ whose image is dense.

Lemma 7.26 (Kuratowski embedding). [Lemma 7.17]

For nonempty X , fix $a \in X$, and define $\Phi(x)(t) = d(x, t) - d(a, t)$ ($t \in X$).

Then $\Phi(x) \in B(X)$, $\|\Phi(x)\|_\infty = d(x, a)$, and $\|\Phi(x) - \Phi(y)\|_\infty = d(x, y)$. As a function of t , $\Phi(x)$ is 2-Lipschitz, so $\Phi(x) \in C_b(X)$. Thus $\overline{\Phi(X)} \subset B(X)$ is a completion of X .

Theorem 7.27 (Existence and uniqueness of completions). [Th. 7.18 and subsequent construction][Ex. 7.45–46]

Every metric space has a completion. For two completions $(\widehat{X}_1, i_1)$, $(\widehat{X}_2, i_2)$, there is a unique onto isometry U with $U \circ i_1 = i_2$. A dense subspace has the same completion as its ambient space. If sets S, T are equipotent, composition with a bijection induces an onto linear isometry $B(S) \cong B(T)$.

Definition 7.28 (Completion by Cauchy classes). Let $\mathcal{C}(X)$ be the set of Cauchy sequences in X , and define $(x_n) \approx (y_n) \iff d(x_n, y_n) \rightarrow 0$, $\widehat{X} = \mathcal{C}(X)/\approx$, $\widehat{d}([(x_n)], [(y_n)]) = \lim_n d(x_n, y_n)$, $i(x) = [(x, \dots)]$.

This is a well-defined complete metric space with dense isometric copy $i(X)$.

Proposition 7.29 (Completion of a normed space). The completion of a normed space has a unique Banach-space structure extending the original structure. In Cauchy-class notation, $\alpha[(x_n)] + \beta[(y_n)] = [(\alpha x_n + \beta y_n)]$, $\|[(x_n)]\| = \lim_n \|x_n\|$. The original space embeds as a dense linear subspace.

Definition 7.30 (Uniform continuity and uniform homeomorphism). [Ex. 7.47, 51]

$f : (X, d) \rightarrow (Y, \rho)$ is uniformly continuous iff

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x, y \in X : d(x, y) < \delta \implies \rho(f(x), f(y)) < \varepsilon.$$

A uniform homeomorphism is a bijection for which both directions are uniformly continuous. Lipschitz maps are uniformly continuous.

Proposition 7.31 (Cauchy sequences and uniform maps). [Ex. 7.25, 48–51]

Uniformly continuous maps preserve Cauchy sequences; uniform homeomorphisms preserve completeness. Distance-to-point and distance-to-nonempty-set maps are uniformly continuous. Every Lipschitz $\mathbb{Q} \rightarrow \mathbb{R}$ extends uniquely to a continuous map on $\mathbb{R}$, preserving its Lipschitz constant. Every continuous $\mathbb{R} \rightarrow \mathbb{R}$ preserves Cauchy sequences, since its domain is complete. For incomplete domains this may fail: $x \mapsto 1/x$ on $(0, 1)$.

8. Compactness

Compact spaces and equivalent criteria

Definition 8.1 (Compactness). A metric space is compact iff it is complete and totally bounded. A subset is compact iff it is compact with its induced metric.

Theorem 8.2 (Sequential compactness). [Th. 8.2][Ex. 8.9]

$$\begin{aligned} X \text{ compact} &\iff \text{every sequence has a subsequence converging in } X \\ &\iff \text{every infinite subset has a limit point in } X. \end{aligned}$$

Definition 8.3 (Covers and the finite intersection property). [Lemma 8.8]

An open cover of X is a family $\mathcal{U} \subset \tau_d$ with $X = \bigcup_{U \in \mathcal{U}} U$. A subcover is a subfamily with the same union. A family $\mathcal{F}$ of subsets has the finite intersection property (FIP) iff every finite subfamily has nonempty intersection.

Theorem 8.4 (Covering and intersection criteria). [Lemma 8.8; Th. 8.9; Cors. 8.10–8.11][Ex. 8.30–32]

For a metric space X , the following are equivalent:

- (i) X is compact.
- (ii) Every open cover has a finite subcover.
- (iii) Every family of closed sets with FIP has nonempty intersection.
- (iv) Every decreasing sequence of nonempty closed subsets has nonempty intersection.
- (v) Every countable open cover has a finite subcover.

Every decreasing sequence of nonempty compact sets in any metric space has nonempty intersection, without an ambient completeness assumption.

Proposition 8.5 (Compact subspaces and operations). [Cor. 8.3][Ex. 8.4,6,11,34]

Compact subsets of a metric space are closed and bounded. Closed subsets of compact spaces are compact. Finite unions and finite products of compact sets are compact. Compactness of a metric subspace is independent of its ambient space. For any $A \subset X$, A closed in $X \iff A \cap K$ compact for every compact $K \subset X$.

Theorem 8.6 (Heine–Borel and finite-dimensional examples). [Ex. 8.1–2,7–8,10]

$K \subset \mathbb{R}^k$: K compact $\iff K$ closed and bounded.

Every finite-dimensional norm sphere and closed bounded ball are compact. For nonempty compact $K \subset \mathbb{R}$, $\inf K, \sup K \in K$. Every compact subset of $\mathbb{R}^k$ is contained in a product of compact intervals. Relative closedness and boundedness do not suffice in $\mathbb{Q}$: $\{q \in \mathbb{Q} : 2 \leq q^2 \leq 3\}$ is not compact. For an Archimedean ordered field F , the following are equivalent: least-upper-bound completeness; compactness of every closed bounded subset of F ; and the existence, for every bounded sequence in F , of a subsequence converging to a point of F .

Proposition 8.7 (Precompactness). [Ex. 8.12–19]

In a complete metric space, A totally bounded $\iff \overline{A}$ compact.

Equivalently X is totally bounded iff its completion is compact. For a compact X , separability and a countable open base follow. A metric space is separable iff it is homeomorphic to a totally bounded metric space, specifically to a subspace of H . For $c_n > 0$,

$$K_c = \{x \in \ell^2 : |x_n| \leq c_n \ \forall n\} \text{ compact} \iff \sum_n c_n^2 < \infty.$$

In particular $\{x \in \ell^2 : |x_n| \leq 1/n\}$ is compact. The Hilbert cube (H, d_H) is compact. Here “precompact” means having compact closure; in a complete ambient space this is exactly total boundedness.

Example 8.8 (Compactness versus properness). [Ex. 8.5]

Every compact space has compact closed balls, but the converse fails for $\mathbb{R}$. A metric space with compact closed bounded balls is called proper. The unit ball of ℓ^∞ is closed and bounded but not compact. A subset of a discrete metric space is compact iff it is finite.

8. Compactness

Continuous functions on compact spaces (continued)

Theorem 8.9 (Continuous images and extrema). [Th. 8.4; Cors. 8.5–8.7][Ex. 8.3,21–23]

Continuous images of compact sets are compact. On nonempty compact X , every continuous $f : X \rightarrow \mathbb{R}$ is bounded and attains both extrema:

$$\exists x_-, x_+ \in X \forall x \in X : f(x_-) \leq f(x) \leq f(x_+).$$

Thus $\|f\|_\infty = \max_X |f|$ is a norm on $C(X)$. For continuous $f : [a, b] \rightarrow \mathbb{R}$, $f([a, b]) = [\min f, \max f]$. For nonempty compact $K \subset X$, its diameter is attained: $\text{diam } K = d(x, y)$ for some $x, y \in K$. Every continuous map with compact domain is closed; every continuous injective map with compact domain is an embedding. Compactness is invariant under homeomorphisms.

Proposition 8.10 (Continuous witnesses to noncompactness). [Ex. 8.20]

For noncompact $E \subset \mathbb{R}$, there exists an unbounded continuous $g : E \rightarrow \mathbb{R}$, and a bounded continuous $h : E \rightarrow \mathbb{R}$ with no maximum. One may use a nonnegative unbounded g and $h = g/(1 + g)$.

Proposition 8.11 (Graphs, segments, and injective curves). [Ex. 8.24–27]

For $f : [a, b] \rightarrow \mathbb{R}$, f continuous $\iff x \mapsto (x, f(x))$ continuous $\iff \text{graph}(f)$ compact in $\mathbb{R}^2$.

Every nondegenerate line segment in a normed space is homeomorphic to $[0, 1]$. A continuous injective $[0, 1] \rightarrow [0, 1]^2$ has nowhere dense image in the square. Every continuous open $\mathbb{R} \rightarrow \mathbb{R}$ is strictly monotone.

Proposition 8.12 (Compact fixed-point results). [Ex. 8.28–29]

Every continuous $[a, b] \rightarrow [a, b]$ has a fixed point. If nonempty compact X and $T : X \rightarrow X$ satisfy $d(Tx, Ty) < d(x, y)$ ($x \neq y$), then T has exactly one fixed point.

Lemma 8.13 (Nested sets are eventually inside a neighborhood). [Ex. 8.33]

If X is compact, F_n decrease through nonempty closed sets, and $\bigcap_n F_n \subset U$ for open U , then $F_n \subset U$ eventually.

Lemma 8.14 (Lebesgue number). [Ex. 8.35]

Every open cover $\mathcal{U}$ of a nonempty compact metric space admits $\delta > 0$ such that every subset with diameter $< \delta$ lies in some $U \in \mathcal{U}$.

Proposition 8.15 (Positive separation and semicontinuous extrema). [Ex. 8.36–39]

If $K \neq \emptyset$ is compact and $F \neq \emptyset$ closed with $K \cap F = \emptyset$, then $d(K, F) > 0$. On a nonempty compact space, every real lower semicontinuous function is bounded below and attains its minimum; every upper semicontinuous function is bounded above and attains its maximum.

Proposition 8.16 (Rigidity of compact self-maps). [Ex. 8.40–43]

On compact X , any isometry $X \rightarrow X$ is onto. More generally, $d(Tx, Ty) \geq d(x, y)$ ($x, y \in X$) $\implies T$ is an onto isometry.

Every onto map satisfying $d(Tx, Ty) \leq d(x, y)$ is likewise an isometry. Outside compact spaces, the right shift on c_0 is an isometry into a proper subset.

Proposition 8.17 (Uniform continuity criteria). [Ex. 8.44–48,56]

Uniform continuity of $f : X \rightarrow Y$ is equivalent to either condition: $d(x_n, y_n) \rightarrow 0 \implies \rho(f(x_n), f(y_n)) \rightarrow 0$,

$$\forall \varepsilon > 0 \exists \delta > 0 \forall A \subset X : \text{diam } A < \delta \implies \text{diam } f(A) < \varepsilon.$$

Uniformly continuous maps preserve Cauchy sequences. Every map from a uniformly discrete space (in particular usual $\mathbb{N}$) is uniformly continuous.

Theorem 8.18 (Heine–Cantor and total boundedness). [Prop. 8.14; Th. 8.15][Ex. 8.69]

Uniformly continuous maps send totally bounded sets to totally bounded sets. Every continuous map from a compact metric space to any metric space is uniformly continuous.

Proposition 8.19 (Algebraic operations and gluing). [Ex. 8.49–50]

Finite sums of uniformly continuous real functions are uniformly continuous. Products are uniformly continuous if both factors are bounded; without boundedness the product can fail, as $x \cdot x = x^2$ on $\mathbb{R}$. If f is uniformly continuous on $(0, 2)$ and $(1, 3)$, it is uniformly continuous on $(0, 3)$. Uniform continuity on every $[n, n + 1]$, $n \in \mathbb{Z}$, does not suffice for uniform continuity on $\mathbb{R}$, again by $f(x) = x^2$.

8. Compactness

Uniform continuity and extension (continued)

Theorem 8.20 (Extension from a dense subspace). [Th. 8.16; Cor. 8.17][Ex. 8.51,65–66,71]

Let $D \subset X$ be dense, Y complete, and $f : D \rightarrow Y$ uniformly continuous. There is a unique continuous extension $F : X \rightarrow Y$, which is uniformly continuous: $F(x) = \lim_n f(x_n)$ whenever $x_n \in D$, $x_n \rightarrow x$.

If f is an isometry or L -Lipschitz, F has the same property. If f is a uniform homeomorphism, it extends to an onto uniform homeomorphism between the completions. Dense subspaces have the same completion; completions are unique up to the unique isometry fixing the original space. For continuous $f : (0, 1) \rightarrow \mathbb{R}$, f uniformly continuous $\iff f(0+), f(1-)$ both exist in $\mathbb{R}$. In this case f extends continuously to $[0, 1]$ and is bounded.

Proposition 8.21 (Further uniform continuity facts). [Ex. 8.52–55]

If f is continuous on a neighborhood of a , then

$$g_a(x) = \frac{f(x) - f(a)}{x - a} \quad (x \neq a),$$

$$f'(a) \text{ exists in } \mathbb{R} \iff g_a \text{ is uniformly continuous on some punctured neighborhood of } a.$$

A continuous real function with finite limits at both infinities is uniformly continuous; in particular this holds when both limits are 0. If bounded $E \subset \mathbb{R}$ is noncompact, it admits a continuous real function that is not uniformly continuous. The bounded continuous function $\sin(x^2)$ on $\mathbb{R}$ is not uniformly continuous; the unbounded function x is uniformly continuous.

Definition 8.22 (Hölder continuity). [Ex. 8.57–60]

$f : X \rightarrow Y$ is α -Hölder with constant L iff $\rho(f(x), f(y)) \leq Ld(x, y)^\alpha$ ($x, y \in X$), $\alpha > 0$.

Every such map is uniformly continuous. On a real interval, an α -Hölder map with $\alpha > 1$ is constant. A real function with bounded derivative on an interval is Lipschitz. For $0 < \alpha \leq 1$, $|s^\alpha - t^\alpha| \leq |s - t|^\alpha$ ($s, t \geq 0$).

For $\alpha > 0$, x^α on $(0, \infty)$ is uniformly continuous iff $\alpha \leq 1$. The exponent $\alpha = 0$ separately gives a constant function. $\sqrt{x}$ is uniformly continuous on $[0, \infty)$ but not Lipschitz there; $1/x$ on $(0, 1)$ is continuous but not uniformly continuous.

Proposition 8.23 (Sequence-space maps and embeddings). [Ex. 8.67–68,70,72]

The diagonal map $D : \ell^2 \rightarrow \ell^1$, $(Dx)_n = x_n/n$, is Lipschitz with constant $(\sum n^{-2})^{1/2}$. Coordinatewise multiplication by a fixed $y \in \ell^\infty$ is Lipschitz on ℓ^1 . If D_0 is dense in X , then X embeds isometrically into $B(D_0)$. Thus every separable metric space embeds isometrically into ℓ^∞ . The closed complete subset $K = \{x \in \ell^\infty : x_n \rightarrow 1\}$

is invariant under the right shift S , and $S|_K$ is an isometry with no fixed point.

Definition 8.24 (Equivalence, uniform equivalence, strong equivalence). [Ex. 8.61–62]

Metrics d, ρ on X are equivalent iff their identity maps in both directions are continuous; uniformly equivalent iff both are uniformly continuous; strongly equivalent iff

$$\exists c, C > 0 \forall x, y \in X : \quad cd(x, y) \leq \rho(x, y) \leq Cd(x, y).$$

strong equivalence $\implies$ uniform equivalence $\implies$ equivalence.

Uniform equivalence preserves completeness and total boundedness; strong equivalence also preserves boundedness.

Example 8.25 (Strictness of the implications). [Ex. 8.63–64,73–74]

d and $d/(1+d)$ are uniformly equivalent; if d is unbounded, they are not strongly equivalent. On $[0, 1]$, $|x - y|$ and $\sqrt{|x - y|}$ are uniformly equivalent but not strongly equivalent. For $x, y \in \mathbb{R}^2$,

$$\rho(x, y) = \frac{\|x - y\|_2}{\sqrt{1 + \|x\|_2^2} \sqrt{1 + \|y\|_2^2}}$$

is a metric equivalent to Euclidean distance but not uniformly equivalent to it. For example, $x_n = (n, 0)$, $y_n = (n + 1, 0)$ have $\rho(x_n, y_n) \rightarrow 0$ but Euclidean distance 1.

Proposition 8.26 (Compact-domain equivalence). [Prop. 8.19]

If (X, d) is compact, another metric ρ on X is equivalent to d iff it is uniformly equivalent to d .

8. Compactness

Bounded linear maps and finite-dimensional normed spaces (continued)

Theorem 8.27 (Continuity criteria for linear maps). [Th. 8.20][Ex. 8.75]

For linear $T : V \rightarrow W$ between normed vector spaces, the following are equivalent: Lipschitz continuity; uniform continuity; continuity everywhere; continuity at 0; continuity at one point; and

$$\exists C < \infty \forall x \in V : \|Tx\|_W \leq C\|x\|_V.$$

They are also equivalent to boundedness of T on the unit ball, or to mapping every bounded set to a bounded set. Such a linear map is called bounded. A nonzero linear map has unbounded range.

Definition 8.28 (Operator norm and the dual space). [Ex. 8.79,82–84]

Let $\mathcal{L}(V, W)$ be the space of bounded linear maps (the textbook's $B(V, W)$). Its norm is

$$\|T\| = \sup_{\|x\| \leq 1} \|Tx\| = \inf\{C \geq 0 : \|Tx\| \leq C\|x\| \forall x\}.$$

If $V \neq \{0\}$, this also equals $\sup_{x \neq 0} \|Tx\|/\|x\| = \sup_{\|x\|=1} \|Tx\|$. For $V = \{0\}$, $\|T\| = 0$. $\|S + T\| \leq \|S\| + \|T\|$, $\|\lambda T\| = |\lambda|\|T\|$, $\|ST\| \leq \|S\|\|T\|$.

If W is Banach, $\mathcal{L}(V, W)$ is Banach, even when V is incomplete. In particular the dual $V^* = \mathcal{L}(V, \mathbb{F})$ is always Banach.

Example 8.29 (Operator norms). [Ex. 8.76–78,80–81]

T	Domain and codomain	$\ T\ $
$x \mapsto \langle x, y \rangle$	$\mathbb{F}^k \rightarrow \mathbb{F} \ (\ \cdot\ _2)$	$\ y\ _2$
$x \mapsto x_j$	$\ell^\infty \rightarrow \mathbb{F}$	1
$x \mapsto (x_n/n)$	$\ell^2 \rightarrow \ell^1$	$\left(\sum_{n \geq 1} n^{-2}\right)^{1/2}$
$f \mapsto \int_a^b f(t) dt$	$C([a, b]) \rightarrow \mathbb{R}$	$b - a$
$f \mapsto (x \mapsto \int_a^x f(t) dt)$	$C([a, b]) \rightarrow C([a, b])$	$b - a$

The function-space norms in the last two rows are supremum norms.

Theorem 8.30 (Equivalent norms). [Cor. 8.21]

For two norms on a vector space,

$$\begin{aligned} \text{equivalent} &\iff \text{uniformly equivalent} \\ &\iff \text{strongly equivalent} \\ &\iff \exists c, C > 0 : c\|x\|_1 \leq \|x\|_2 \leq C\|x\|_1 \quad (\forall x). \end{aligned}$$

Theorem 8.31 (Finite-dimensional structure). [Th. 8.22; Cors. 8.23–8.26][Ex. 8.85–88]

All norms on a finite-dimensional vector space are equivalent. Every linear map with finite-dimensional domain is bounded. Every finite-dimensional normed space is complete, and every finite-dimensional subspace of any normed space is closed. For $\dim V = k$ and a basis $v_1, \dots, v_k$,

$$\|(a_1, \dots, a_k)\|_V = \left\| \sum_{j=1}^k a_j v_j \right\|_V$$

defines a norm on $\mathbb{F}^k$ making the coordinate map an onto linear isometry. Any two normed spaces of the same finite dimension over the same scalar field are linearly homeomorphic, with both maps Lipschitz.

Proposition 8.32 (Infinite-dimensional contrasts). [Ex. 8.89]

For a normed space V , $\dim V < \infty \iff \{x : \|x\| \leq 1\}$ compact.

V is infinite-dimensional iff it admits a discontinuous linear functional, iff it has a proper dense linear subspace (with the usual axiom of choice). For $p < \infty$, c_{00} is a proper dense subspace of ℓ^p . All separable infinite-dimensional Banach spaces are mutually homeomorphic; for $1 \leq p < q < \infty$, ℓ^p, ℓ^q are not uniformly homeomorphic.